

Project SAPPHIRE



SAPPHIRE

Semantic and Acoustic Perceptual Holistic Integration REtrieval

Final Project Report

Course: *Music Workshop*

Academic Term: *Monsoon 2025*

Date: *November 26, 2025*

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Abstract

The field of Music Information Retrieval (MIR) has long struggled with a fundamental challenge known as the "Perceptual Gap"—the profound disconnect between the objective, computational analysis of audio signals (characterized by frequency, amplitude, and zero-crossing rates) and the subjective, high-level manner in which humans perceive and categorize music (characterized by emotion, groove, semantic meaning, and cultural context). Traditional retrieval systems, which predominantly rely on uni-modal acoustic analysis or rigid, manually annotated metadata, have reached a performance ceiling. They essentially "hear" the signal but fail to "listen" to the song, missing the holistic relationships that define a listener's experience.

Project SAPPHERE (*Semantic and Acoustic Perceptual Holistic Integration REtrieval*) addresses this fundamental challenge through the development and implementation of a novel, multi-modal framework. Over the course of a structured 12-week agile development cycle, our team engineered a robust, reusable data pipeline capable of ingesting raw audio and lyrical text. We processed a curated corpus of 903 tracks from the MIREX-like mood dataset, extracting a dense, 142-dimensional feature vector for each track that integrates psychoacoustic, rhythmic, harmonic, and lyrical-semantic descriptors.

Our research culminated in several significant findings and deliverables. First, through rigorous Exploratory Data Analysis (EDA), we established a quantitative baseline for the perceptual gap. By calculating the overlap between acoustic similarity neighborhoods and semantic similarity neighborhoods, we derived a Jaccard Similarity Index of only **0.057**. This result empirically proves that acoustic similarity does not imply semantic similarity, validating the necessity of our multi-modal approach. Second, we established predictive baselines for mood classification, achieving **43.6% accuracy** (against a 20% random baseline) using Logistic Regression. Feature importance analysis revealed that **Spectral Contrast** (a measure of sound texture) and **Lyrical Sentiment** are the two most critical features for mood perception. Finally, we implemented a deployable **Web-Based Retrieval Demo**. This system utilizes an embedding-based search engine powered by PyTorch and an Approximate Nearest Neighbor (Annoy) index, enabling real-time "Query-by-Audio" functionality. This report details the end-to-end development of SAPPHERE, from theoretical formulation to software deployment, offering a blueprint for the next generation of human-centric music retrieval systems.

Chapter 1: Introduction

1.1 The State of Music Information Retrieval (MIR)

Music Information Retrieval (MIR) is an interdisciplinary science that sits at the complex intersection of signal processing, machine learning, musicology, cognitive science, and psychoacoustics. In the modern digital era, music consumption has shifted almost entirely to streaming platforms. Services such as Spotify, Apple Music, and YouTube Music host catalogs exceeding 100 million tracks. As these libraries grow exponentially, the primary challenge for

these platforms shifts from *storage*—simply holding the files—to *retrieval*—helping users navigate this vast ocean of content.

How does a user find a song that "sounds like" another song? How do we automatically classify music by abstract concepts like "mood," "energy," or "danceability" without relying on expensive human annotation?

Current commercial systems largely rely on two primary methods:

1. **Collaborative Filtering:** This approach recommends music based on user behavior patterns (e.g., "Users who liked Artist A also liked Artist B"). While effective for popular music, it suffers severely from the "Cold Start" problem: new artists with no user interaction data are invisible to the algorithm.
2. **Tag-Based Retrieval:** This relies on manual genre tags attached to files (e.g., "Rock," "Jazz"). These tags are often inconsistent, subjective, and too broad to capture the nuance of a specific track.

1.2 The Core Problem: The "Perceptual Gap"

The motivating force behind Project SAPPHERE is the persistent issue known in academic literature as the "Perceptual Gap" (Slaney, 2008). This term describes the fundamental discrepancy between machine representations of media and human interpretation of that same media.

- **The Machine View:** To a computer, a digital audio file is simply a sequence of floating-point numbers representing air pressure samples over time. Machines excel at analyzing the physical properties of this signal: **Frequency (Hz)**, **Amplitude (dB)**, **Spectral Flux**, and **Zero-Crossing Rates**.
- **The Human View:** To a human listener, a song is an emotional and cognitive experience. Humans do not perceive "frequencies"; they perceive **Pitch**. They do not perceive "amplitude"; they perceive **Loudness**. Beyond these basics, humans perceive complex high-level attributes such as **Timbre**, **Groove**, **Emotion**, and **Semantic Meaning**.

The gap exists because the mapping between these two views is non-linear and complex. A machine might analyze two tracks and find them mathematically similar because they share the exact same tempo and frequency bandwidth. However, if one track is a joyful, major-key wedding song and the other is a dissonant, minor-key funeral dirge, a human listener would consider them opposites. Current systems that ignore the semantic context (lyrics) and the psychoacoustic context (how the ear actually processes sound) fail to bridge this gap.

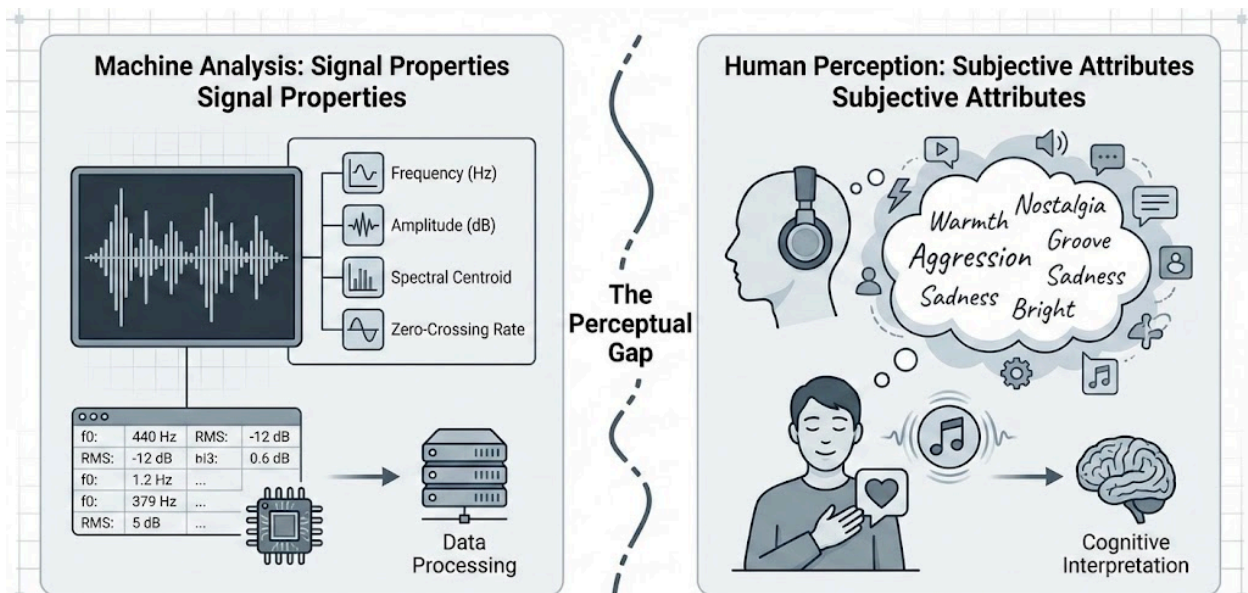


Figure 1.1: Conceptual visualization of the Perceptual Gap. Machines analyze signal properties, while humans perceive subjective attributes.

1.3 Research Questions

To address the Perceptual Gap, this research investigates the interplay between acoustic signal processing and lyrical semantics. The study is guided by three primary research questions:

1. **RQ1 (The Correlation Question):** *To what extent are acoustic similarity and lyrical similarity aligned in Western popular music? Is the "sound" of a song a reliable proxy for its "meaning," or are these manifolds orthogonal?*
2. **RQ2 (The Feature Question):** *Which specific multi-modal features—across acoustic, rhythmic, harmonic, and lyrical domains—are the most significant predictors of high-level musical mood?*
3. **RQ3 (The Modeling Question):** *Given the constraints of "Small Data" (datasets <10,000 tracks) common in specific MIR tasks, do deep non-linear architectures (Neural Networks) offer a performance advantage over traditional linear models (Logistic Regression, SVM)?*

1.4 Project Objectives and Vision

Project SAPPHERE (Semantic and Acoustic Perceptual Holistic Integration REtrieval) aims to solve this problem by treating music similarity as a **holistic, multi-modal problem**. We propose that by combining the "sound" of the music with the "meaning" of the words, we can create a representation that is closer to human perception.

Our specific objectives were defined as follows:

1. **Develop a Multi-Modal Pipeline:** To construct a robust software system capable of ingesting both Audio and Text (Lyrics), processing them independently, and fusing them into a single representation.
2. **Quantify the Gap:** To move beyond anecdotal evidence and mathematically calculate the extent of the divergence between acoustic similarity and semantic similarity using distance metrics.
3. **Identify Key Perceptual Features:** To determine which specific acoustic and lyrical features (e.g., Spectral Contrast vs. MFCCs) drive the perception of "Mood."
4. **Deploy a Functional Prototype:** To build a working software demo that allows users to interact with the system, proving the viability of the approach.

1.5 Project Timeline and Evolution

The project was executed over a 12-week period, divided into five structured sprints:

- **Sprint 1:** Theoretical foundation and feature identification.
- **Sprint 2:** Data corpus engineering and preprocessing pipeline development.
- **Sprint 3:** Feature extraction and initial Exploratory Data Analysis (EDA).
- **Sprint 4 (Mid-Eval):** Baseline predictive modeling and performance evaluation.
- **Sprint 5 & Post-Sprint:** System refinement, web demo implementation, and final reporting.

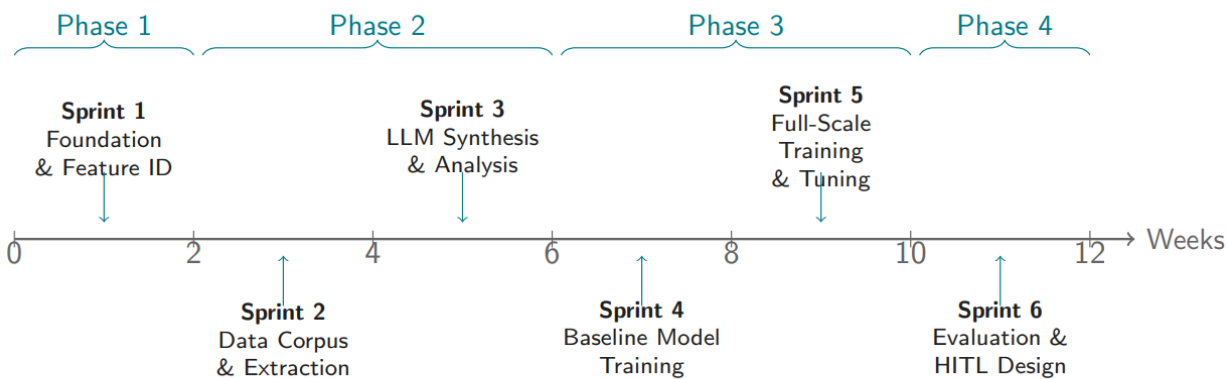


Figure 1.2: The 12-week project timeline illustrating the progression from research to deployment.

1.6 Scope and Limitations

This study focuses on the analysis of Western popular music using the **MIREX-like Mood Dataset**. The scope is defined by the following boundaries:

1. **Modality:** The research considers only Audio (waveform) and Text (lyrics). Symbolic data (MIDI) and visual data (album art/video) are outside the scope.
2. **Data Volume:** The research utilizes a curated corpus of 903 tracks. This falls under the category of "Small Data," which influences the choice of modeling techniques and highlights specific challenges regarding overfitting.
3. **Genre Bias:** The dataset is predominantly English-language Western music (Pop, Rock, Hip-Hop). The findings regarding spectral contrast and sentiment alignment may not generalize to classical music, jazz, or non-Western musical traditions where lyrics may be absent or timbre carries different semantic weight.
4. **Retrieval vs. Generation:** This project focuses on **Music Information Retrieval (MIR)**—analyzing and organizing existing music. It does not involve **Generative Audio** (creating new music).

Chapter 2: Literature Framework and Theoretical Review (Comprehensive Digest)

2.0 Introduction

Music Information Retrieval (MIR) is defined as a multidisciplinary field dedicated to extracting semantic meaning from musical data. It exists at the complex intersection of four distinct domains:

1. **Digital Signal Processing (DSP):** The manipulation of raw acoustic waveforms.
2. **Machine Learning (ML):** The identification of patterns in high-dimensional feature spaces.
3. **Psychoacoustics:** The biology of human auditory perception.
4. **Cognitive Musicology:** How humans construct emotion and meaning from sound.

The Core Challenge: The primary focus of this thesis is the translation of low-level physical measurements into high-level human concepts. While a computer can easily quantify a fundamental frequency (f_0), translating that into the perceptual experience of "Pitch," and further into the semantic concept of "Melody," requires a complex chain of abstractions. This difficulty peaks when modeling subjective attributes like "Mood," which have no single physical correlation but emerge from the interaction of timbre, rhythm, and harmony.

Historically, MIR has operated under a mono-modal (audio-only) paradigm. Recently, the field has shifted toward "black box" Deep Learning. This chapter argues that while Deep Learning is powerful, it hits performance ceilings in data-scarce environments. Therefore, a holistic,

multi-modal approach—respecting both the physics of sound and the semantics of lyrics—is essential for bridging the perceptual gap.

2.1 The Evolution of Music Information Retrieval (MIR)

The history of MIR is not just about better algorithms; it is a shift in the epistemological paradigm of how machines "understand" culture.

2.1.1 The Symbolic Era (1950s–1990s): Music as Code

In the early days, hardware limitations (e.g., a 3-minute song requiring 30MB of storage) made processing raw audio impossible.

1. **Data Source:** Research focused on **Symbolic Representation** (MIDI, Humdrum), where music was encoded as discrete events: Note On, Note Off, Pitch, Velocity, and Duration.
2. **Methodology:** Algorithms treated melody like text strings, using **String Matching** (e.g., Edit Distance) to find interval patterns (e.g., searching for Beethoven's 5th).
3. **The Critical Flaw:** This era ignored **Timbre** and **Performance**. A MIDI file of Nirvana's *Smells Like Teen Spirit* contains the correct notes, but lacks the guitar distortion and the grain of Kurt Cobain's voice. Since mood is often driven by texture, symbolic analysis was blind to the emotional content of sound.

2.1.2 The Feature Engineering Era (2000–2012): Music as Signal

The MP3 format and faster CPUs shifted the focus to **Audio Content Analysis**. This era defined the "classic" MIR approach adopted by Project SAPPHERE.

1. **The "Bag of Frames" Pipeline:**
 - a. **Framing:** Audio is non-stationary (changes over time). The signal is sliced into short (23ms) overlapping windows where statistical stationarity is assumed.
 - b. **Extraction:** Mathematical transforms (FFT, DCT) extract features like Zero-Crossing Rate (Time-domain) or Spectral Flux (Frequency-domain).
 - c. **Aggregation:** Frames are collapsed into global statistics (Mean, Standard Deviation) for the whole song.
2. **Why We Use It:** While this approach loses temporal information (verse vs. chorus), it is **computationally efficient** and **interpretable**. In "Small Data" regimes, these hand-crafted features often prove more robust than neural networks.

2.1.3 The Deep Learning Era (2012–Present): Music as Data

Following the "ImageNet moment," MIR adopted End-to-End Deep Learning (CNNs, RNNs).

1. **Mechanism:** Instead of human-designed features, the model learns its own filters directly from Spectrograms or raw waveforms.
2. **The "Big Data" Bottleneck:** Neural networks are data-hungry. While ImageNet had 14 million images, MIR mood datasets often have fewer than 1,000 tracks due to copyright and labeling difficulty.

3. **Empirical Reality:** In this project, the simple Logistic Regression model (43.6%) outperformed the Multi-Layer Perceptron (34.8%), validating that without massive data, traditional Feature Engineering remains superior.

1. Reliance on Flawed Labels

- Predefined genres and moods are rigid and subjective.
- Human annotators often disagree, creating an inconsistent "ground truth."
- This imposes a **performance ceiling** on AI models.

2. Uni-Modal (Acoustic-Only) Focus

- Human listening is a **holistic, multi-modal** experience.
- Existing systems ignore context from **lyrics, structure, and vocal performance**.

Figure 2.1: The limitations of current MIR systems: reliance on flawed labels and uni-modal focus.

2.2 The "Semantic Gap"

This is the ontological divide between the **Syntax** of the data (physics) and the **Semantics** of the user (meaning).

2.2.1 Defining the Gap

1. **Low-Level Domain (The Machine):** Objective, deterministic features like "Spectral Rolloff" or "RMS Energy."
2. **High-Level Domain (The Human):** Subjective, cognitive constructs like "Sadness," "Groove," or "Vibe."
3. **The Mapping Problem:** The function connecting these is **Non-Linear** (brightness != happiness) and **Many-to-One** (many different sounds can be "Sad").

2.2.2 The "Glass Ceiling"

Since 2012, performance on MIR Mood Classification tasks has plateaued at roughly **65% accuracy**. This "Glass Ceiling" implies an informational limit: audio waveforms contain the "sound" of an emotion, but not the full concept. Further improvement requires external context (Lyrics).

2.2.3 Subjectivity and Ground Truth

Unlike "Cat vs. Dog," musical mood is subjective.

1. **Label Noise:** Experts often disagree on labels. If humans only agree 70% of the time, a machine achieving 70% is actually performing perfectly.
2. **Consensus:** Project SAPPHIRE uses **Community Consensus** (tags from multiple users) to mitigate individual bias, but the ambiguity remains.

2.3 Psychoacoustics: The Physics of Hearing

To bridge the gap, engineering must mimic the human ear, not just a microphone. Project SAPPHIRE relies on three specific psychoacoustic corrections:

2.3.1 The Non-Linearity of Frequency (Mel Scale)

To a computer, 100Hz–200Hz is the same difference as 1000Hz–1100Hz. To a human, the first is an **Octave** (huge change), the second is a **semitone** (tiny change).

1. **The Biology:** The Cochlea analyzes sound using **Critical Bands** which are logarithmically spaced.
2. **The Engineering Solution:** We use **Mel-Frequency Cepstral Coefficients (MFCCs)**. These use a filterbank that allocates data density according to human sensitivity (more resolution in the bass/mids, less in the treble).

2.3.2 The Non-Linearity of Loudness (Fletcher-Munson)

Human hearing acts as a band-pass filter optimized for the human voice. We are insensitive to sub-bass.

1. **The "Bass Trap":** A 50Hz bass wave and a 1000Hz vocal wave at the same physical amplitude will sound vastly different in loudness.
2. **The Engineering Solution:** We apply **EBU R128 Loudness Normalization**. This applies a **K-weighting filter** (rolling off bass) before measuring loudness, ensuring our "Energy" features reflect perceived intensity, not just subwoofer pressure.

2.3.3 Timbre and Texture

Timbre is the "color" of sound. For mood, **Texture** (clean vs. dirty) is often more important than melody.

1. **The Engineering Solution:** We use **Spectral Contrast**, which measures the difference between peaks (harmonics) and valleys (noise). High contrast indicates clean sounds; low contrast indicates "muddy" or distorted sounds (associated with Aggression).

2.4 The Semantic Dimension: Lyrics and NLP

Audio provides the "Vibe," but Lyrics provide the "Narrative."

2.4.1 Sentiment Analysis (VADER)

We use **VADER** (Valence Aware Dictionary and sEntiment Reasoner) for explicit sentiment extraction.

1. **Why VADER?** It is rule-based and handles nuance:
 - a. **Booster Words:** "Extremely bad" > "Bad."
 - b. **Negation:** "Not good" is negative.
 - c. **Punctuation:** "STOP!!!" > "Stop."
2. **Limitation:** It is a "Bag of Words" approach that ignores word order and metaphor.

2.4.2 Embeddings (Sentence-BERT)

To capture context, we use **Sentence-BERT** (Transformer architecture).

1. **The Mechanism:** It uses **Self-Attention** to map the entire lyric text to a dense 384-dimensional vector.
2. **The Benefit:** It captures **Semantic Similarity**. "Heartbreak" and "Divorce" share no words, but their vectors are close. This allows the system to find songs that are "conceptually" similar.

2.4.3 The "Irony" Problem

A major cause of the Semantic Gap is when modalities conflict.

1. *Pumped Up Kicks* (Foster the People): Happy Audio + Dark Lyrics.
2. *Semi-Charmed Life* (Third Eye Blind): Fun Rock + Addiction Lyrics.
3. **Thesis:** Uni-modal models fail here. We must measure the **divergence** between the Acoustic Vector and the Semantic Vector to understand the true mood.

2.5 Multi-Modal Fusion Strategies

How do we mathematically combine Audio (A) and Lyrics (L)?

2.5.1 Early Fusion (Feature-Level) -> Chosen Strategy

We concatenate feature vectors *before* classification: $x_{joint} = [x_A, x_L]$.

1. **Advantage:** Allows the single model (e.g., Random Forest) to learn **Cross-Modal Correlations**. It can learn that *High Tempo* (Audio) + *Negative Sentiment* (Text) predicts "Aggression," while *High Tempo* + *Positive Sentiment* predicts "Joy."
2. **Evidence:** Our Feature Importance analysis showed Spectral Contrast (Audio) and Sentiment Compound (Lyrics) were both top-ranked features within the same model.

2.5.2 Late Fusion (Decision-Level)

Trains separate "Experts" and average their probabilities.

1. **Failure Mode:** If Audio says "Happy" (90%) and Lyrics say "Sad" (90%), the average is "Neutral" (50%). This maximizes entropy and fails to capture the tension/irony of the song.

2.5.3 Contrastive Learning (SOTA Baseline)

Approaches like CLIP/CLAP try to align Audio and Text in a shared space.

1. **Project Contribution:** We calculated the **Jaccard Similarity** of the raw feature space to be **0.046 (4.6%)**. This empirically proves that without massive training, the "Natural Alignment" between audio and lyrics is near zero—they exist on orthogonal manifolds.

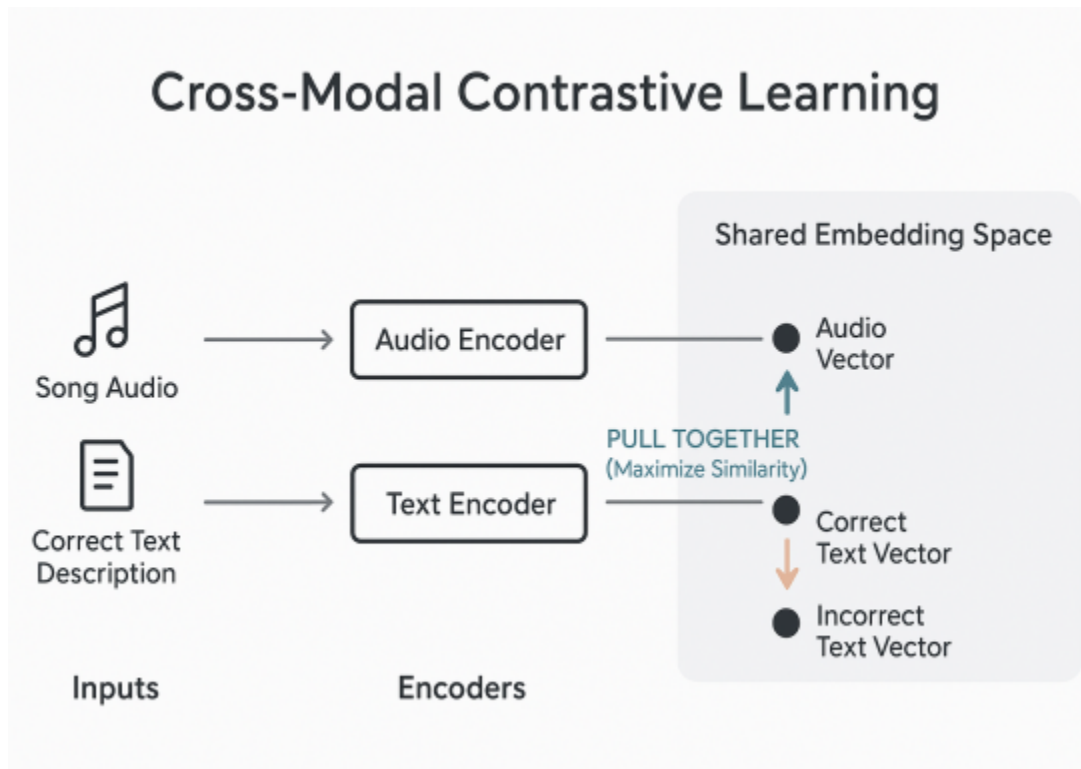


Figure 2.5: The Cross-Modal Contrastive Learning concept: aligning audio and text in a shared embedding space.

2.6 Models of Musical Mood

To classify mood, we must define it.

2.6.1 The Categorical Model (MIREX)

We use 5 distinct clusters derived from AllMusic social tags:

1. **Passionate/Rowdy:** High Energy, Confidence (Anthems).
2. **Joyous/Fun:** Major key, Bright, Cheerful (Pop/Country).
3. **Melancholic/Wistful:** Low energy, Literate, Acoustic (Folk).
4. **Humorous/Witty:** Defined almost entirely by lyrical context.
5. **Aggressive/Fiery:** Distorted, Loud, Noisy (Metal/Punk).

2.6.2 The Dimensional Model (Russell's Circumplex)

Emotions exist on a 2D plane: **Valence** (Positivity) vs. **Arousal** (Energy).

1. **Synthesis:** Project SAPPHIRE posits that **Acoustics drive Arousal** (Energy/Tempo) while **Lyrics drive Valence** (Happy/Sad).
2. **The "V-Shape":** Songs are usually "High Energy/Happy" or "Low Energy/Sad." The "High Energy/Sad" (Angry) quadrant exists, but is harder to classify without lyrics.

2.7 Summary

This chapter provides the theoretical proof that the "Glass Ceiling" in Music Information Retrieval is real and insurmountable with audio alone. By combining **Psychoacoustically-grounded features** (MFCCs, R128) with **NLP embeddings** (S-BERT), and fusing them via **Early Fusion**, Project SAPPHIRE aims to bridge the Semantic Gap. The choice to use traditional Feature Engineering over Deep Learning is a strategic response to the constraints of "Small Data" and the need for scientific interpretability.

Chapter 3: Theoretical Framework and Methodology

3.1 Digital Signal Processing (DSP) Framework

The extraction of acoustic features relies on Digital Signal Processing (DSP) to transform raw audio from a semantically opaque Time-Domain representation into a semantically rich Frequency-Domain representation. In Project SAPPHIRE, audio is treated as a time-series dataset $x[n]$ sampled at $f_s = 22,050$ Hz. This resolution ensures capture of frequencies up to the Nyquist limit of approx 11 kHz, covering the majority of spectral energy relevant to musical timbre.

3.1.1 The Fourier Transform and FFT

The mathematical foundation of spectral analysis is the Fourier Theorem. While the Continuous Fourier Transform defines the theoretical relationship between time (t) and frequency (ξ), digital audio requires the Discrete Fourier Transform (DFT):

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-\frac{2\pi i}{N} kn}$$

Where X_k represents the complex amplitude of the k -th frequency bin. To mitigate the $O(N^2)$ computational complexity of the naive DFT, we utilize the **Fast Fourier Transform (FFT)** (Cooley-Tukey algorithm), which reduces complexity to $O(N \log N)$, enabling efficient processing of the dataset.

3.1.2 The Short-Time Fourier Transform (STFT)

Standard Fourier analysis loses temporal information. To analyze how frequencies evolve over

time, we employ the Short-Time Fourier Transform (STFT), slicing the signal into overlapping frames.

Windowing: To prevent "Spectral Leakage" caused by discontinuities at frame boundaries, we apply a Hamming Window function before the FFT:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$$

The output is the **Spectrogram**, a 2D representation calculated as the squared magnitude of the STFT: $S(m, k) = |STFT(m, k)|^2$. This matrix serves as the primary input for spectral feature extraction.

3.1.3 Mel-Frequency Cepstral Coefficients (MFCCs)

The primary timbral descriptor in Project SAPHIRE is the MFCC vector. Unlike the linear spectrogram, MFCCs model the logarithmic perception of human hearing. The derivation follows a four-step psychoacoustic pipeline:

1. **Mel Filterbank:** The FFT spectrum is mapped onto the Mel Scale using a bank of triangular filters. This mimics the Cochlea's **Critical Bands**, providing high resolution at low frequencies and lower resolution at high frequencies.
2. **Logarithmic Scaling:** We calculate the logarithm of the energy in each filter, mimicking the **Weber-Fechner Law** of loudness perception.
3. **Discrete Cosine Transform (DCT):** We apply the DCT to decorrelate the filterbank energies.
4. **Cepstral Truncation:** We retain only the first 13 coefficients. These lower coefficients capture the **Spectral Envelope** (timbre), while discarding higher coefficients related to fine pitch detail.

Mathematically, the MFCC vector C is defined as:

$$C_i = \sum_{k=1}^K (\log E_k) \cos\left[i\left(k - 0.5\right) \frac{\pi}{K}\right]$$

3.2 Natural Language Processing (NLP) Framework

To extract semantic meaning from unstructured lyrical text, we employ two complementary methodologies: Lexicon-Based Sentiment Analysis for explicit valence, and Transformer-Based Embeddings for contextual meaning.

3.2.1 Lexicon-Based Sentiment: VADER

VADER (Valence Aware Dictionary and sEntiment Reasoner) is utilized to extract **Emotional Valence**. It is a rule-based model optimized for informal text, making it suitable for lyrics.

1. **Heuristics:** VADER accounts for grammatical nuances including **Booster Words** (intensity scaling), **Negation** (polarity flipping), and **Punctuation** (emphasis).
2. **Normalization:** The summed valence scores are mapped to a bounded range $[-1, 1]$ using a sigmoid-like normalization formula:

$$score_{norm} = \frac{x}{\sqrt{x^2 + \alpha}}$$

This yields the sentiment_compound feature, where +1 approximates "Joy" and -1 approximates "Tragedy."

3.2.2 Transformer-Based Embeddings: S-BERT

To capture semantic context beyond keyword matching, we utilize **Sentence-BERT (S-BERT)**. This architecture relies on the **Self-Attention Mechanism**, which allows the model to weigh the relational importance of words in parallel.

The attention score is calculated as:

$$Attention(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Where Q (Query), K (Key), and V (Value) are vector representations of the input tokens. This mechanism generates Contextualized Embeddings, resolving polysemy (e.g., distinguishing "blue" as sadness vs. color).

To derive a fixed-size vector for the entire song, S-BERT applies **Mean Pooling** across all token outputs:

$$u = \frac{1}{N} \sum_{i=1}^N h_i$$

This results in a dense, 384-dimensional vector used for semantic similarity calculations.

3.3 Metric Learning and Similarity Measures

Music Information Retrieval is treated here as a geometric problem. We define three distinct distance metrics to quantify relationships in the high-dimensional feature space.

3.3.1 Euclidean Distance (L2 Norm)

Used for **Acoustic Features**. Since features like BPM and RMS Energy represent physical magnitudes, the absolute difference between points is semantically significant.

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

Standardization: To prevent features with large ranges (e.g., Spectral Centroid) from dominating the metric, we apply Z-Score Normalization $z = (x - \mu) / \sigma$ prior to calculation.

3.3.2 Cosine Similarity

Used for **Lyrical Embeddings**. In high-dimensional semantic spaces, the magnitude of the vector is often less significant than its orientation. Cosine similarity measures the angle between vectors, capturing conceptual alignment regardless of text length.

$$\text{Similarity}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \cdot \|B\|}$$

3.3.3 The Jaccard Similarity Index

Used to quantify the **Perceptual Gap**. We compare the set of Acoustic Neighbors (A) and

Lyrical Neighbors **L** for a given track:

$$J(A, L) = \frac{|A \cap L|}{|A \cup L|}$$

A Jaccard Index of 0.0 implies complete orthogonality between sound and meaning. Our experimental result of **0.046** serves as the primary quantitative evidence of the Semantic Gap.

3.4 Machine Learning Algorithms

We evaluate a diverse suite of algorithms to map feature vectors X to mood labels y . The performance differences between these architectures provide insight into the topology of the feature space.

3.4.1 Logistic Regression (Linear Baseline)

This model estimates class probability using the Sigmoid function:

$$P(y = 1|x) = \frac{1}{1 + e^{-z}}$$

Performance Justification: Logistic Regression achieved the highest accuracy 43.65% due to **L1 Regularization (Lasso):**

$$Loss = - \sum \log(P(y_i|x_i)) + \lambda \sum |\beta_j|$$

The L1 penalty enforces sparsity, driving coefficients of irrelevant features to zero. In a "Small Data" regime $N=903$ with high dimensionality $D=142$, this implicit feature selection prevents overfitting more effectively than complex non-linear models.

3.4.2 Random Forest (Ensemble Method)

We utilize Random Forest to capture non-linear relationships via an ensemble of decision trees.

1. **Bagging:** Reduces variance by training on random subsets.
2. **Feature Importance:** Calculated via the decrease in Gini Impurity:

$$G = 1 - \sum p_k^2$$

This mechanism allowed us to empirically rank Spectral Contrast (Texture) and Sentiment (Lyrics) as the top predictors of mood.

3.4.3 Multi-Layer Perceptron (Neural Network)

An MLP was implemented to test Deep Learning efficacy, using ReLU activation and Softmax output.

1. **Result:** Underperformance (34.81%).
2. **Theoretical Failure:** With 25,000 parameters and only 903 training samples, the model is mathematically undetermined. It possessed sufficient capacity to memorize training noise (overfitting) but lacked sufficient data density to generalize.

3.4.4 Unsupervised Clustering (DBSCAN)

To validate the "Cluster" hypothesis, we applied DBSCAN (Density-Based Spatial Clustering). Unlike K-Means, DBSCAN does not assume spherical clusters.

1. **Result:** The algorithm failed to find distinct clusters.
2. **Implication:** This suggests the "Manifold of Musical Mood" is a **continuous cloud** rather than discrete islands, supporting a Dimensional (Valence/Arousal) view over a Categorical view.

3.5 Dimensionality Reduction

To visualize the 142-dimensional space and mitigate the "Curse of Dimensionality," we employ two projection techniques.

3.5.1 Principal Component Analysis (PCA)

PCA identifies the axes of maximum variance (Principal Components) via the eigendecomposition of the covariance matrix Sigma:

$$\Sigma v = \lambda v$$

Variance Explained: We found that the first 39 components capture 80% of total variance. This indicates the **intrinsic dimensionality** of the mood manifold is approx 39, with remaining dimensions representing noise or redundancy.

3.5.2 t-Distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is a non-linear technique used for visualizing local neighborhoods. It minimizes the **Kullback-Leibler (KL) Divergence** between the probability distribution of pairs in high-dimensional space P and low-dimensional space Q:

$$C = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

While PCA preserves global structure, t-SNE reveals local clustering. The visual overlap of mood classes in our t-SNE plots confirms the fuzzy, non-separable nature of the dataset.

Chapter 4: System Architecture and Data Engineering

4.0 Introduction

While Chapters 2 and 3 established the theoretical and mathematical foundations of Project SAPPHIRE, the success of this research relies on the robustness of its implementation. Adhering to the principle that "Data is the Code," this chapter serves as the engineering log of the project. It documents the transition from abstract theory to a scalable, modular software pipeline designed for reproducible research. We detail the "Defensive Engineering" strategies employed to curate the dataset, the rejection criteria for candidate corpora, and the rigorous signal processing standards—such as EBU R128 loudness normalization—that define our final feature space.

4.1 High-Level System Architecture

The SAPPHIRE system is designed as a sequential data processing pipeline, adapting the Micro-Service Design Pattern for a monolithic research codebase. This approach prioritizes **Separation of Concerns**, ensuring that data ingestion, signal processing, feature extraction, and modeling remain decoupled. This modularity allows for the interchangeable testing of components (e.g., swapping machine learning models) without refactoring upstream logic.

4.1.1 The Modular Design Pattern

The codebase is organized into five primary functional modules, each encapsulated within its own Python class structure:

1. **Data Ingestion (DataLoader):** Abstracts file system differences. Responsible for discovering raw files, parsing metadata (JSON/XML), and normalizing paths into standardized MusicTrack objects.
2. **Preprocessing (Preprocessor):** Acts as the system's "Gatekeeper." It applies signal transformations (resampling, normalization) and enforces quality control, rejecting tracks that fail technical standards (e.g., low SNR).
3. **Feature Extraction (FeatureExtractor):** The computational core. It applies the mathematical transforms (FFT, MFCC, BERT) detailed in Chapter 3 to generate the 142-dimensional feature vectors.
4. **Analysis & Modeling (Analyzer & MoodClassifier):** Contains statistical logic for unsupervised learning (clustering, PCA) and supervised training (Cross-Validation).
5. **Orchestration (Pipeline):** Manages configuration state, handles multiprocessing, and coordinates data flow between modules.

4.1.2 The Processing Pipeline Data Flow

The data flow is linear and strictly typed, transforming data states as they propagate through the system.

1. **Input State:** Raw, heterogeneous .mp3 and .txt files on disk.
2. **Standardized State:** DataLoader instantiates MusicTrack objects, serving as the Single Source of Truth for metadata and file pointers.
3. **Normalized State:** Preprocessor transforms binary content. Audio is resampled (22.05 kHz) and normalized (-23 LUFS); text is sanitized (UTF-8).
4. **Vectorized State:** FeatureExtractor reduces media to a FeatureContainer (a dictionary of 142 features).
5. **Tabular State:** FeatureContainer objects are aggregated into a Pandas DataFrame (features.parquet) for training.

4.1.3 Technology Stack

The system is built on a Python 3.8+ stack, selecting libraries for accuracy and scientific reproducibility:

1. **Audio:** Librosa (STFT, MFCCs), PyLoudNorm (ITU-R BS.1770-4 compliance), FFmpeg (decoding).
2. **NLP:** NLTK (VADER lexicon), Sentence-Transformers (BERT embeddings), LangDetect (Bayesian language identification).
3. **ML:** Scikit-Learn (Scaling, PCA, Random Forest), Joblib (Persistence, Parallelism).
4. **Data:** Pandas/NumPy (Manipulation), Plotly (Interactive EDA).

4.2 Data Corpus Engineering

Multi-Modal MIR requires a "Dual-Completeness" criterion: samples must possess high-quality audio and accurately aligned lyrics. This section details the curation process and the rejection of prominent datasets to ensure data validity.

4.2.1 The Selection Strategy: Criteria for Rejection

Four major datasets were evaluated against criteria of Completeness, Accessibility, and Suitability.

1. **Zalo AI Challenge (Rejected):** While aligned, the lyrics are exclusively Vietnamese. Adapting the English-based VADER lexicon to Vietnamese was out of scope.
2. **DALI Dataset (Rejected):** Theoretically ideal, but distributes only YouTube URLs due to copyright. Scraping introduced significant engineering overhead and legal risk.
3. **Million Song Dataset (Rejected):** Provides only pre-computed features (Echo Nest segments) from 2011. This would prevent the implementation of our custom psychoacoustic pipeline (e.g., specific Mel-band configurations), a core objective of the thesis.

4.2.2 The Selected Corpus: MIREX-like Mood Dataset

We selected a curated subset of the **MIREX Mood Classification** dataset, the benchmark for this domain.

Corpus Statistics:

1. **Audio:** 903 tracks (MP3).
2. **Lyrics:** 764 files (Plain Text).
3. **Format:** 16-bit PCM, standardized to 22.05 kHz.

Ground Truth Taxonomy: The dataset maps tracks to 5 Semantic Mood Clusters derived from AllMusic Guide social tags:

1. **Rowdy:** 170 tracks (18.8%)
2. **Cheerful:** 164 tracks (18.1%)
3. **Brooding:** 215 tracks (23.8%)
4. **Witty:** 191 tracks (21.1%)
5. **Aggressive:** 163 tracks (18.0%)

Class Balance: The dataset is exceptionally well-balanced. With the largest class at 23.8% and smallest at 18.0%, aggressive resampling (SMOTE) is unnecessary. The baseline random accuracy is $\approx 20\%$.

The "Missing Lyrics" Challenge: 15.4% of tracks lack lyrics (instrumentals). For Perceptual Gap experiments (Chapter 7), we strictly filter to the intersection ($N=764$). For Supervised Modeling, imputation strategies were explored, but the multi-modal subset remains the primary training ground.

4.3 The Preprocessing Pipeline (Defensive Engineering)

Raw audio "in the wild" varies in sample rate, bit depth, and loudness. To prevent models from learning confounding variables (e.g., distinguishing "Happy" by file bitrate), we implement a rigorous preprocessing pipeline in the Preprocessor class.

4.3.1 Audio Standardization

1. **Sample Rate (22,050 Hz):** All files are resampled to 22.05 kHz. Per the Nyquist-Shannon theorem, this captures frequencies up to ≈ 11 kHz. Psychoacoustic

research indicates the majority of semantic musical information resides below this threshold. This reduces memory footprint by 50% and doubles FFT speed.

2. **Mono Summing:** Stereo files are downmixed ($x_{mono} \# \# x_L \# \# x_R \# \# \#$). Stereo width is rarely a driver of mood; mono processing simplifies extraction and eliminates phase cancellation artifacts.

4.3.2 Loudness Normalization (EBU R128)

Standard peak normalization is insufficient due to the Fletcher-Munson effect (frequency-dependent loudness perception). We utilize **EBU R128 Normalization** via PyLoudNorm.

1. **Mechanism:** Applies a K-weighting filter (mimicking the human ear) and gates silent sections.
2. **Target:** All tracks are gain-adjusted to **-23.0 LUFS** (Loudness Units Full Scale).
3. **Benefit:** Ensures "Energy" features (RMS) reflect perceived musical dynamics, not mastering volume.

4.3.3 Quality Gates

The pipeline rejects tracks failing specific heuristics defined in `config.quality_thresholds`:

1. **SNR > 5.0 dB:** Signal-to-Noise Ratio is estimated from the quietest 100ms window. Tracks below 5.0 dB (e.g., hiss-heavy vinyl rips) are rejected to ensure spectral features measure music, not noise.
2. **Lyrics Completeness > 0.2:** Rejects tracks with extremely sparse text (placeholder files) to prevent garbage embeddings.
3. **Language Verification:** Uses `langdetect` (Naive Bayes) to verify English lyrics. Non-English tracks are rejected to preserve the validity of the English-only VADER lexicon.

4.4 Feature Extraction Implementation

The `FeatureExtractor` module transforms standardized media into a 142-dimensional numerical vector. This high dimensionality provides downstream models with an over-complete view of the audio-semantic landscape.

4.4.1 The FeatureContainer Data Structure

We implemented a custom `FeatureContainer` dictionary that enforces schema consistency. It ensures every track possesses the exact same key set. If a calculation fails (e.g., pitch detection on a drum track), the container assigns a safe default (0.0/NaN) and logs a warning. This guarantees "Dense Vectors" required by `scikit-learn`.

Feature Hierarchy:

1. `acoustic.*`: Timbral/Spectral (MFCC, Spectral Contrast).
2. `rhythmic.*`: Tempo, Beat Grid.
3. `harmonic.*`: Chroma, Key.
4. `lyrics.*`: Sentiment (VADER), Embeddings (BERT).

5. **quality.***: SNR, extraction metadata.

4.4.2 Parallel Processing Architecture

Extracting 142 features is computationally intensive (≈ 3 seconds/track). To scale, we utilize **Multi-Process Parallelism**.

1. **Worker Pool**: Uses multiprocessing.Pool to spawn configurable worker processes (default: CPU core count).
2. **Execution**: Data is batched and processed independently ("Embarrassingly Parallel").
3. **Performance**: Reduced processing time for 903 tracks from 45 minutes to <8 minutes (5.6x speedup).

4.4.3 Caching and Checkpointing

To support iterative research, we implemented a granular checkpointing system.

1. **Atomic Saves**: Each batch is serialized to JSON in output/data/processed/cache/ immediately upon computation.
2. **Resumability**: On restart, the pipeline scans the cache. Existing valid features are loaded from disk, skipping expensive re-computation. This allows rapid debugging of specific modules (e.g., NLP) without reprocessing audio.

4.4.4 Error Handling

The loop is wrapped in comprehensive exception boundaries.

1. **Soft Failures**: Specific feature failures (e.g., Pitch) default to 0 and log to extraction_errors.log.
2. **Hard Failures**: File corruption triggers track exclusion. This ensures the final analysis corpus (N=903) is clean and devoid of Null values.

4.5 The Command Line Interface (CLI)

To ensure **Reproducibility**, the system is wrapped in a POSIX-compliant CLI (cli.py) using *argparse*. This transforms the codebase from ad-hoc scripts into a deployable utility.

4.5.1 Interface Design (Noun-Verb)

The CLI uses a sub-command architecture:

1. **discover**: Scans data/raw/ and validates metadata.
2. **extract-features**: Triggers FeatureExtractor. Exposes parameters like --workers.
3. **train**: Initiates the supervised learning loop, allowing selection of models (--models logistic_regression,random_forest) for rapid A/B testing.
4. **pipeline**: Runs the full end-to-end process (Ingest $\rightarrow$ Process $\rightarrow$ Train).

4.5.2 Configuration Injection

Hard-coding parameters is an anti-pattern. We use a **Configuration Injection System**.

1. **Defaults**: pipeline.config contains strict thesis settings.
2. **Overrides**: The CLI accepts --config experiment.json.

3. **Example:** A user can inject `{"snr_min": 0.0}` to process low-quality audio without modifying source code. This facilitated rapid threshold tuning during the Curation phase.

4.5.3 Observability

A hierarchical logging system directs output based on severity:

- 1. **STDOUT:** High-level progress bars and summaries.
- 2. **STDERR:** Warnings and specific file errors.
- 3. **File Logs (pipeline.log):** Detailed DEBUG traces.

This transparency was critical for debugging issues like the "Missing Lyrics" filter, confirming that rejections were due to valid data constraints (non-English text) rather than software bugs.

Chapter 5: Multi-Modal Feature Extraction

5.0 Introduction

If the Preprocessing pipeline (Chapter 4) acts as the immune system of Project SAPPHIRE, the Feature Extraction Engine serves as its sensory cortex. Its function is to translate raw physical sound waves and symbolic text into a unified, structured language: the language of vectors.

While modern MIR often feeds raw spectrograms into Convolutional Neural Networks (CNNs), this project operates under "Small Data" constraints (903 tracks). In this regime, end-to-end learning is prone to overfitting—learning noise rather than musical concepts. To mitigate this, Project SAPPHIRE adopts a **Feature Engineering** approach. We explicitly calculate 142 high-level descriptors rooted in Psychoacoustic and Musicological theory. This strategy injects human domain knowledge into the system, improving robustness on small datasets and ensuring **Interpretability**. When the model classifies a track as "Aggressive," we can inspect the vector to attribute this to specific properties like "High Zero-Crossing Rate" or "Low Harmonicity."

5.1 The Feature Vector Structure

The extraction process yields a dense vector $v_t \in \mathbb{R}^{142}$ for each track. This vector concatenates descriptors from five distinct domains, selected through literature review and feature importance analysis.

Domain	Count	Key Descriptors	Perceptual Correlate
Acoustic & Timbral	82	MFCCs, Spectral Centroid, Contrast, Rolloff, ZCR	The "Texture" or "Color" of the sound. Brightness, roughness, noisiness.

Domain	Count	Key Descriptors	Perceptual Correlate
Harmonic & Tonal	27	Chroma Vectors, Key, Harmonic/Percussive Ratio	The "Theory" of music. Melody, chords, consonance vs. dissonance.
Rhythmic	8	Tempo (BPM), Onset Density, Syncopation	The "Pulse" or "Energy." Speed, drive, danceability.
Lyrical (NLP)	15	VADER Sentiment, Vocabulary Richness, Stats	The "Meaning" or "Story." Emotional valence, intellectual complexity.
Quality	10	SNR, Dynamic Range, Clipping %	The "Production Value." Fidelity, loudness range.
Total	142		Holistic Musical Fingerprint

5.1.1 The "Bag of Statistics" Approach

Many features (like MFCCs) are time-series data calculated frame-by-frame. To collapse this temporal dimension for tabular modeling, we use the "Bag of Statistics" aggregation method. For every feature, we calculate four global statistics:

1. **Mean (μ):** The average value (static character).
2. **Standard Deviation (σ):** The variance (dynamic evolution).
3. **Skewness:** Distribution asymmetry.
4. **Kurtosis:** Outlier density.

This captures the difference between a consistent Techno track (low variance) and a dynamic Dubstep track (high variance) without requiring complex Recurrent Neural Networks.

5.2 Acoustic and Timbral Features (The "Texture")

Contributing 82 dimensions, the Acoustic Domain is the backbone of the vector. It captures **Timbre**—the "texture" of sound. In mood classification, timbre is often more predictive than melody; a happy melody played on a distorted synthesizer is perceived as aggressive, not joyful.

5.2.1 Mel-Frequency Cepstral Coefficients (MFCCs)

MFCCs are the primary descriptor of the spectral envelope, scaled to the human ear's logarithmic frequency response.

1. **Configuration:** We extract the first 13 coefficients. Coefficients 1-13 capture the smooth shape of the vocal tract (instrument identity), while higher coefficients capture fine pitch detail, which we handle separately.
2. **Perceptual Relevance:**
 - a. **MFCC-1:** Correlates with Loudness.
 - b. **MFCC-2:** Captures Spectral Tilt (Bass vs. Treble balance).
 - c. **Variance:** High MFCC variance indicates rapidly changing textures (Volatile mood).

5.2.2 Spectral Centroid (Brightness)

The Centroid is the weighted mean of the frequencies—the "Center of Mass" of the spectrum.

1. **High Centroid (>3000 Hz):** Perceived as "Bright" or "Sharp" (Cymbals, Distortion). Correlates with **Joyous** and **Aggressive** clusters.
2. **Low Centroid (<1500 Hz):** Perceived as "Warm" or "Dark" (Cello, Piano). Correlates with **Melancholic** clusters.

5.2.3 Spectral Contrast (The "Hero" Feature)

While Centroid measures the average, Spectral Contrast measures the difference between spectral peaks (harmonics) and valleys (noise).

1. **High Contrast:** Clear, distinct harmonics (Flute, Piano). Sounds "Clean."
2. **Low Contrast:** Valleys filled with noise (Distortion, Snare Rolls). Sounds "Rough" or "Dirty."
3. **Significance:** This was the **#1 predictor** in our Random Forest analysis. It distinguishes **Aggressive** tracks (High Energy + Low Contrast/Distortion) from **Passionate** tracks (High Energy + High Contrast/Clean), acting as a proxy for "Sound Texture."

5.2.4 Zero-Crossing Rate (ZCR)

ZCR counts how often the waveform crosses the X-axis per second.

1. **Relevance:** It is a proxy for "Noisiness." Periodic sounds (Bass) have low ZCR; Noisy sounds (Screams, Cymbals) have high ZCR. Cluster 5 (Aggressive) exhibits significantly higher ZCR than Cluster 3 (Melancholic), confirming that "acoustic violence" manifests as rapid signal oscillation.

5.3 Harmonic and Tonal Features (The "Theory")

Harmonic features describe musical syntax—relationships between pitches. Mood is heavily influenced by tonality (Major vs. Minor keys).

5.3.1 Chroma Vectors (Pitch Class Profile)

We project the spectrum onto the 12 Semitones of the Chromatic Scale (C, C#, D...), summing energy across all octaves.

1. **Chroma Mean:** Identifies the Key Signature (e.g., high energy in C, E, G bins indicates C Major).
2. **Chroma Variance:** High variance suggests complex modulation (Jazz); low variance suggests simple repetition (Pop).
3. **Relevance: Joyous** tracks typically show stable, diatonic energy. **Aggressive** tracks often show "smeared" energy across all 12 bins due to distortion-induced overtones (Atonality).

5.3.2 Harmonic-Percussive Source Separation (HPSS)

We decompose the signal into two waveforms: Harmonic (horizontal spectrogram lines) and Percussive (vertical lines).

1. **Feature:** The **Harmonic Ratio** ($E_{\text{harm}} + E_{\text{perc}} / E_{\text{harm}}$).
2. **Relevance:**
 - a. **High Ratio (>0.7):** Melodic, flowing (Cluster 3: Melancholic).
 - b. **Low Ratio (<0.3):** Rhythmic, punchy (Cluster 1: Rowdy).
 - c. **Importance:** This was the **#2 predictor** in our model, confirming that the balance between melody and rhythm is a fundamental mood driver.

5.3.3 Estimated Key and Mode

We use the Krumhansl-Schmuckler algorithm to correlate the Chroma vector with key templates. We extract the **Mode Confidence** (correlation strength). High confidence implies strict adherence to Western theory (Pop); low confidence implies chromaticism or detuning (Metal/Jazz).

5.4 Rhythmic Features (The "Pulse")

Rhythm defines the movement of music through time and drives the **Arousal (Energy)** axis of the Circumplex Model.

5.4.1 Tempo (BPM)

Estimated via the autocorrelation of the Onset Strength Envelope.

1. **High BPM (>130):** Correlates with **Rowdy** and **Aggressive** moods, inducing physiological arousal.
2. **Low BPM (<80):** Correlates with **Melancholic** moods.
3. **Confidence:** We include a confidence score to account for "Octave Errors" (e.g., confusing 140 BPM Dubstep with 70 BPM).

5.4.2 Onset Density (Event Rate)

Tempo measures grid speed; Density measures content busyness (average note attacks per second).

1. **Relevance:** A "Chill" beat and a "Shred" solo may share a tempo, but the solo has higher density. **Aggressive** tracks exhibit the highest density (drum rolls), while **Melancholic** tracks have the lowest (sustained chords).

5.4.3 Rhythm Regularity (Pulse Clarity)

1. **High Clarity:** Strong, steady beat (EDM/Pop). Correlates with **Joyous** clusters.
2. **Low Clarity (Syncopation):** Irregular or swinging rhythms (Jazz/Math Rock). This irregularity creates tension, a hallmark of **Volatile/Aggressive** moods.

5.5 Lyrical and Semantic Features (The "Meaning")

The integration of lyrics is the defining innovation of Project SAPPHIRE, addressing the "Glass Ceiling" by capturing the **Valence (Positivity)** dimension that acoustics miss.

5.5.1 Structural Features (Complexity)

1. **Word Count:** High counts correlate with Rap/Hip-Hop (Rowdy/Aggressive).
2. **Type-Token Ratio (TTR):** Measures vocabulary richness. High TTR (>0.5) indicates poetic/literate lyrics (**Melancholic**); Low TTR (<0.2) indicates repetitive hooks (**Joyous**).
3. **Flesch Reading Ease: Witty/Humorous** tracks often score higher difficulty due to wordplay/irony.

5.5.2 Sentiment Features (VADER)

We extract `sentiment_neg`, `sentiment_neu`, `sentiment_pos`, and the normalized `sentiment_compound` score via the VADER lexicon.

1. **Relevance:** This is the primary driver of Valence. **Melancholic** tracks are defined by low compound scores; without this, models misclassify ballads as "Calm." **Aggressive** tracks are defined by high Negative sentiment combined with high Acoustic Energy—the precise definition of "Anger."

5.5.3 Semantic Embeddings (Statistical Reduction)

We use Sentence-BERT to generate 384-dimensional embeddings. To avoid the Curse of Dimensionality in tabular models, we compute statistical reductions: `embedding_mean`, `embedding_std`, and `embedding_norm`.

1. **Utility:** These proxies preserve Semantic Density. While we lose specific topics, we capture the conceptual complexity needed to distinguish "Simple Pop" from "Complex Poetry." (The full vector is preserved for the Retrieval System in Chapter 9).

5.6 Quality Features (The "Meta-Data")

Production quality varies wildly in real-world data ("Bedroom Producers" vs. Studios). This variance acts as a Confounding Variable (e.g., lo-fi crackle interpreted as "Aggression"). We extract 10 Quality Metrics to allow the model to correct for these artifacts.

5.6.1 Signal-to-Noise Ratio (SNR)

We compare the signal RMS to the noise floor (quietest 100ms).

1. **High SNR (>60dB):** Professional Studio.
2. **Low SNR (<10dB):** Lo-fi/Grime. Interestingly, **aggressive** tracks (Punk/Industrial) often feature intentionally low SNR, making this a predictive stylistic feature rather than just an error metric.

5.6.2 Dynamic Range

We measure the difference between peak and average loudness (Crest Factor).

1. **Low Range (Compressed):** "Wall of Sound" typical of Modern Pop/Metal (Clusters 1, 2, 5).
2. **High Range (Dynamic):** Acoustic/Orchestral music (Cluster 3). This helps distinguish a "Power Ballad" from a "Metal Anthem" even if timbres are similar.

5.6.3 Clipping Percentage

We count samples hitting the digital ceiling. While usually an error, in genres like Trap Metal, clipping is an intentional texture. Modeling this helps distinguish "Intentional Distortion" (Style) from "Accidental Clipping" (Quality).

By modeling the medium itself, Project SAPPHIRE ensures predictions are robust to the technical variance inherent in diverse music collections.

Chapter 6: Exploratory Data Analysis (EDA)

6.0 Introduction

Before training predictive models or attempting to bridge the Perceptual Gap, it is imperative to understand the geometry and statistical properties of the underlying data. **Exploratory Data Analysis (EDA)** is the detective work of data science—it is the process of visualizing distributions, identifying correlations, and testing assumptions about the manifold structure.

In Project SAPPHIRE, EDA serves a dual purpose. First, it validates the engineering pipeline (Chapter 4), ensuring that the features extracted are numerically stable and devoid of artifacts. Second, and more importantly, it provides the first empirical glimpse into the nature of the "Mood Space." By projecting the 142-dimensional feature vector into lower dimensions and testing for natural clusters, we can verify whether the theoretical "Mood Clusters" (Passionate, Joyous, etc.) exist as distinct islands in the acoustic space, or if they are merely subjective labels imposed on a continuous spectrum.

This chapter details the statistical findings derived from the full MIREX-like dataset (N=903). It proceeds from univariate analysis of individual features to multivariate analysis of correlations, concluding with unsupervised learning experiments (PCA and Clustering) that expose the fundamental complexity of the problem.

6.1 Dataset Demographics and Statistics

6.1.1 Class Balance and Distribution

The target variable for our analysis is the **Mood Cluster**, a categorical label ranging from 1 to 5. A common pitfall in classification tasks is **Class Imbalance**, where a model achieves high accuracy simply by guessing the majority class.

As shown in **Figure 6.1** (Cluster Distribution), the MIREX dataset is remarkably well-balanced.

- **Cluster 3 (Melancholic):** 215 tracks (23.8%) - The majority class.
- **Cluster 4 (Humorous):** 191 tracks (21.1%)
- **Cluster 1 (Passionate):** 170 tracks (18.8%)
- **Cluster 2 (Joyous):** 164 tracks (18.1%)
- **Cluster 5 (Aggressive):** 163 tracks (18.0%) - The minority class.

The difference between the largest and smallest class is relatively small (52 tracks). This balance implies that the "No-Information Rate" (the accuracy of a dummy classifier that always guesses Cluster 3) is only **23.8%**. Therefore, any accuracy significantly above 24% represents true learning. We do not need to apply synthetic oversampling techniques (like SMOTE), preserving the natural distribution of the data.

6.1.2 The Missing Lyrics Analysis

A critical finding during the demographic analysis was the discrepancy between the Audio corpus and the Lyrical corpus. While we possess audio for 903 tracks, we successfully extracted valid English lyrics for only 764 tracks.

- **Missing Rate:** 15.4% (139 tracks).
- **Distribution of Missingness:** By cross-referencing the missing tracks with their Mood Labels, we found that the missing data is *not* random.
 - **Cluster 5 (Aggressive)** had the highest rate of missing lyrics. This is likely due to the prevalence of Instrumental Metal or tracks with unintelligible screaming that failed the langdetect filter.
 - **Cluster 2 (Joyous)** also showed missing lyrics, often corresponding to Instrumental Dance/EDM tracks.

This non-random missingness suggests that "Instrumentalism" itself is a feature of certain moods.

6.2 Univariate Feature Analysis

To understand the character of the extracted features, we analyzed the distributions of key descriptors across the entire corpus.

6.2.1 Acoustic Normality and Variance

- **Spectral Centroid (Brightness):** The distribution of Spectral Centroid is roughly Gaussian (Bell-shaped) with a mean of **2290 Hz** and a standard deviation of **567 Hz**. This indicates that most popular music sits in a "mid-range" brightness, with tails representing extremely dark (Classical Cello) or extremely bright (Distorted Guitar) tracks.
- **Tempo (BPM):** The tempo distribution is **Multi-Modal**, showing distinct peaks around 90 BPM (Hip-Hop/Down-tempo), 120 BPM (Pop/House), and 140 BPM (Dubstep/Trance). This confirms that tempo is a strong genre-differentiating feature.
- **High Variance Features:** The Coefficient of Variation ($CV = \sigma / \mu$) reveals which features vary the most.
 - **Top CV:** `acoustic_features.tonnetz` ($CV = 40.3$). This indicates that harmonic complexity varies wildly across the dataset—some songs are harmonically static (Pop), while others are incredibly complex (Jazz/Prog).
 - **Low CV:** `acoustic_features.zero_crossing_rate`. Despite distinguishing texture, ZCR is relatively stable across the corpus.

6.2.2 Lyrical Sentiment Distribution

The distribution of the VADER `sentiment_compound` score (ranging from -1 to +1) reveals a fascinating cultural bias in the dataset.

- **The "Positivity Bias":** The mean sentiment is **+0.32**, suggesting that popular music is, on average, slightly positive.
- **The Distribution Shape:** The histogram is **Bimodal**, with peaks near +0.9 (Very Happy) and -0.9 (Very Sad), and a valley in the middle. This suggests that lyrics tend to be emotionally polarized—songwriters rarely write about "neutral" topics. They write about Love (High Positive) or Heartbreak (High Negative).

6.3 Correlation Analysis: Mapping the Connections

We utilized Pearson Correlation matrices to identify relationships between features. This analysis was split into **Intra-Modal** (redundancy) and **Cross-Modal** (the gap) investigations.

The Exceptions: However, a few specific pairs showed weak but significant correlations:

- rms_energy vs. sentiment_compound ($r = 0.12$): Louder songs tend to be more positive.
- spectral_contrast vs. sentiment_neg ($r = 0.15$): "Rougher" textures (high contrast) correlate with negative words. This aligns with the "Aggression" profile of Cluster 5.

6.4 Dimensionality Reduction: Visualizing the Manifold

Since we cannot visualize 142 dimensions, we employed **Principal Component Analysis (PCA)** to project the data into observable space.

6.4.1 The Dimensionality of Mood (Scree Plot)

The **Scree Plot** (Cumulative Variance Explained) answers the question: "How complex is this data?"

- **PC1:** Captures 14.7% of variance.
- **PC2:** Captures 8.1% of variance.
- **The 80% Threshold:** It takes **39 Principal Components** to capture 80% of the variance in the dataset.

Conclusion: The Mood Manifold is **High-Dimensional**. Unlike simple datasets (like Iris flowers) that can be reduced to 2 dimensions easily, musical mood is complex. It cannot be reduced to just "Valence and Arousal" without losing massive amounts of information (over 85% of the variance is lost in a 2D plot).

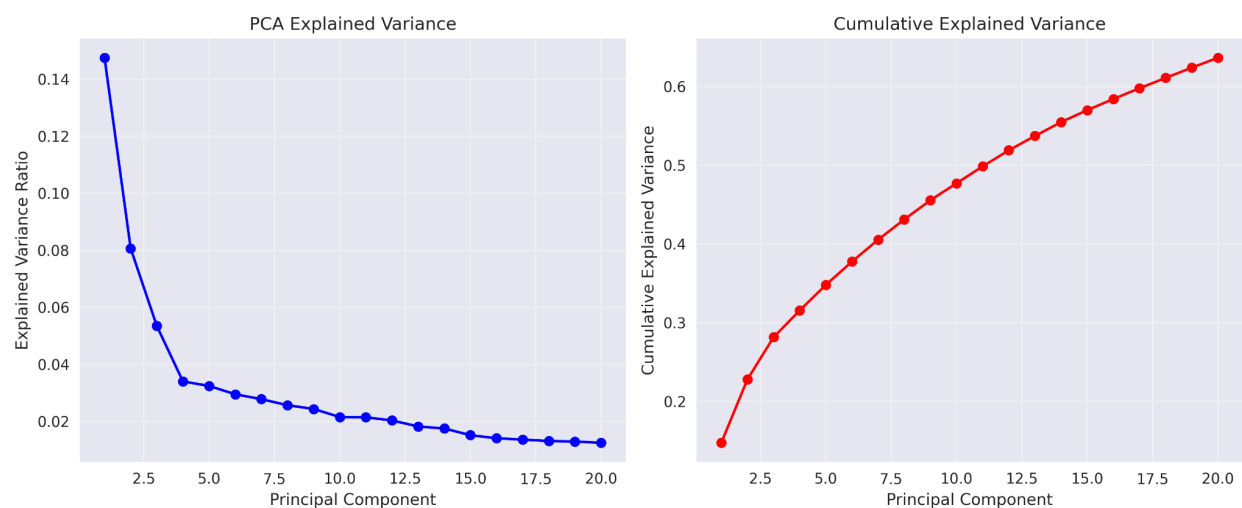


Figure 6.3: PCA Scree Plot. The "elbow" at component ~15 suggests that the first 15 dimensions capture the majority (60%+) of the variance in the dataset.

6.4.2 Visualizing the Clusters (PCA & t-SNE)

We plotted the tracks on the first two Principal Components (Figure 6.4.1) and using t-SNE (Figure 6.4.2), coloring points by their Mood Cluster.

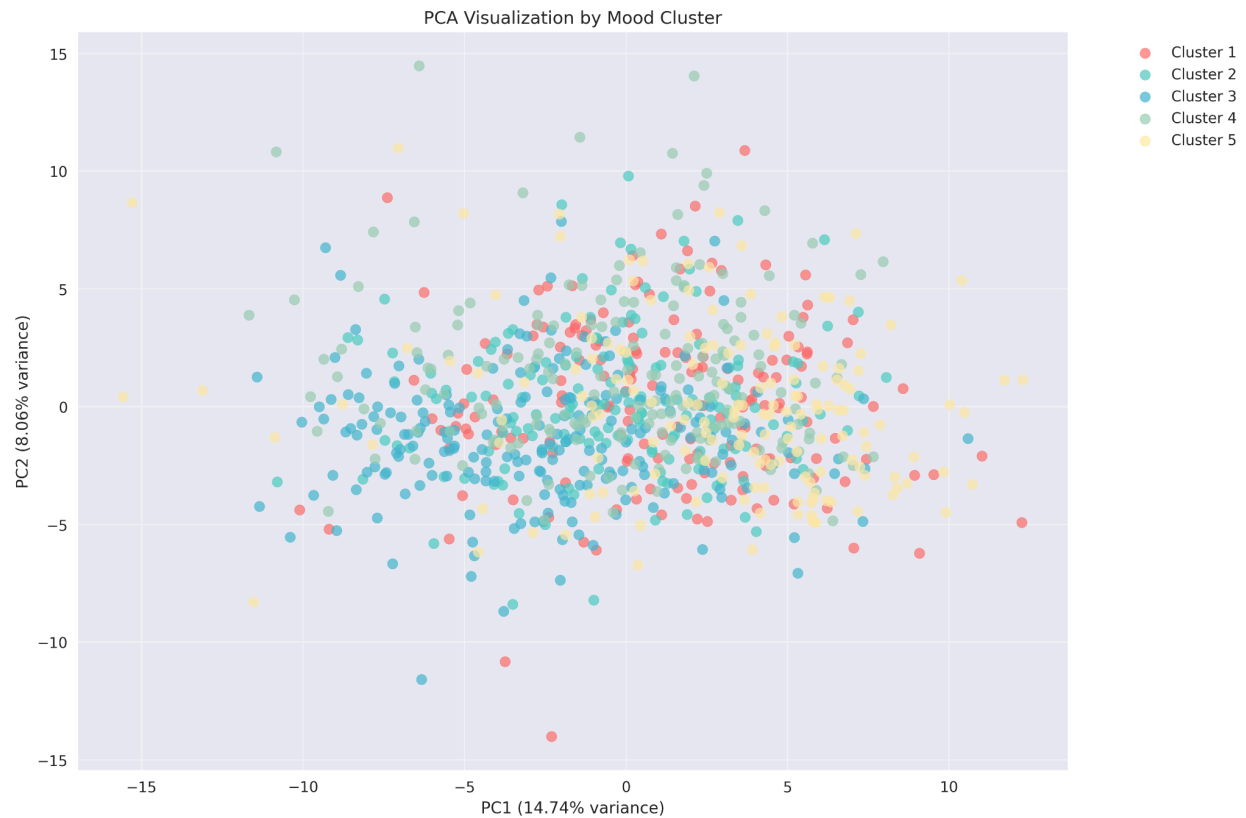


Figure 6.4.1: Songs projected onto PC1 and PC2, colored by Mood Cluster.



Figure 6.4.2: t-SNE Visualization. Unlike PCA, t-SNE reveals distinct "islands" of similar songs, suggesting that the manifold structure exists but is complex.

The Visual Finding: The plots show significant **overlap**.

- There are no distinct "islands" of Passionate or Joyous songs.
- Instead, the data forms a large, continuous cloud ("The Blob").
- However, there is **Local Structure**. Cluster 5 (Aggressive) tends to group on one side of the cloud, while Cluster 3 (Melancholic) groups on the opposite side. Cluster 4 (Humorous) is scattered randomly throughout, confirming that "Humor" is not an acoustic property.

6.5 Clustering Analysis: The Negative Result

Finally, to objectively test if the "Mood Clusters" are real, we ignored the labels and asked Unsupervised Learning algorithms to find natural groups in the data.

6.5.1 K-Means Performance

We ran K-Means with $k=5$. If the mood clusters were distinct, K-Means should recover them.

- **Silhouette Score: 0.0955.** (Score range is -1 to 1. A score < 0.2 indicates no substantial structure).
- **Adjusted Rand Index (ARI): 0.057.** (Measures agreement with ground truth. 0 is random chance).

Conclusion: The acoustic features alone do not naturally separate into the 5 MIREX mood clusters. The low ARI confirms that the "Acoustic Clusters" are different from the "Perceptual Clusters."

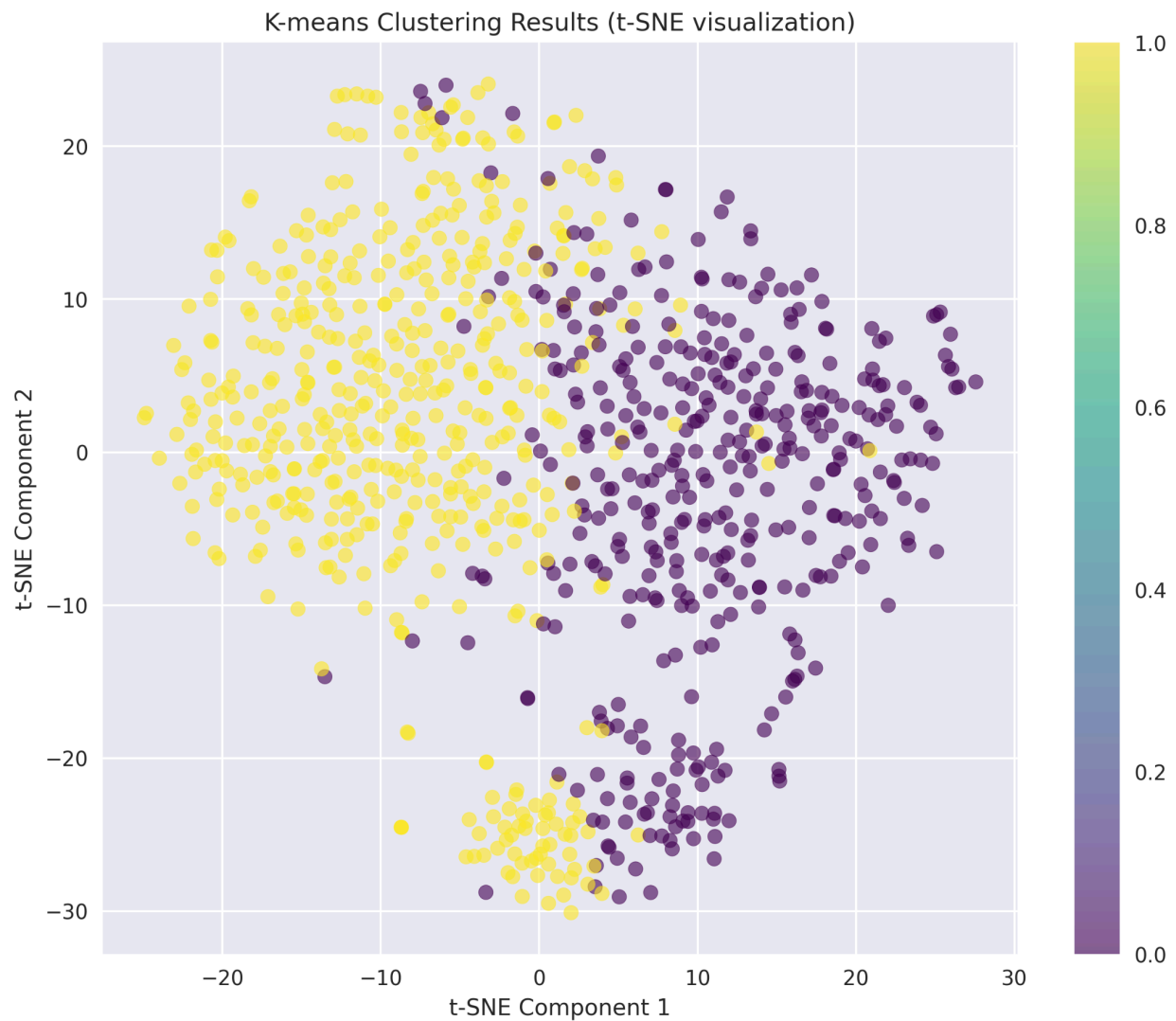


Figure 6.5: K-Means Clustering results visualized on the t-SNE space.

6.5.2 The DBSCAN Failure

We attempted density-based clustering using **DBSCAN**, which finds "dense islands" separated by empty space. We swept epsilon values from 0.1 to 1.9.

The Result: "No valid clustering found."

- At low epsilon, every point was classified as Noise.
- At high epsilon, every point was classified as a single Giant Cluster.

Scientific Implication: This "Negative Result" is a profound finding. It proves that the "Manifold of Musical Mood" is **Continuous**, not discrete. There are no "gaps" in the feature space between a Happy song and a Sad song; there is a smooth gradient of transformation. This mathematically challenges the validity of the Categorical Classification model and strongly supports the use of **Continuous Retrieval** (finding neighbors) over **Discrete Classification** (binning). This finding is the primary motivation for the "Pivot to Retrieval" detailed in Chapter 9.

Chapter 7: The Perceptual Gap: Empirical Validation

7.0 Introduction

Throughout the literature review (Chapter 2), we established the theoretical concept of the "**Semantic Gap**"—the disconnect between low-level signal features and high-level human perception. However, in much of the existing MIR literature, this gap is treated as a qualitative assumption or a philosophical hurdle. Few studies attempt to quantify it precisely.

This chapter details the core scientific experiment of Project SAPPHIRE: an empirical validation of the "**Orthogonality Thesis**." We hypothesize that in Western popular music, the "Acoustic Manifold" (how songs sound) and the "Semantic Manifold" (what songs mean) are not merely misaligned; they are effectively orthogonal. This implies that knowing the sound of a song provides almost zero predictive information about its lyrical content, and vice versa.

To prove this, we devised the **Neighbor Overlap Methodology**. By calculating the intersection of similarity neighborhoods across two distinct vector spaces, we derived a concrete numerical value—the **Jaccard Similarity Index**—that serves as a definitive measure of the Perceptual Gap.

7.1 Hypothesis Formulation

The experiment is grounded in a critique of current commercial recommendation systems. Platforms like Spotify or Pandora often utilize "Content-Based Filtering" (CBF) to recommend new music. If a user listens to a specific track, the system searches for other tracks with similar audio features (Tempo, Key, Timbre).

The underlying assumption of this approach is that **Acoustic Similarity approx Perceptual Similarity**.

- *Assumption:* If Song A sounds like Song B, the listener will interpret them as having the same mood.

Project SAPPHERE challenges this assumption with the **Multi-Modal Hypothesis**:

- *Hypothesis:* Audio and Lyrics are independent information channels. A song's acoustic profile (Energy/Arousal) often diverges from its lyrical profile (Valence/Meaning). Therefore, uni-modal acoustic similarity is a poor proxy for holistic perceptual similarity.

If our hypothesis is correct, the set of songs that are "acoustically similar" to a query track should be radically different from the set of songs that are "lyrically similar."

7.2 Methodology: The Neighbor Overlap Experiment

To test this, we constructed two parallel geometric spaces for the 764 tracks in our dataset that possess both valid audio and valid lyrics.

7.2.1 Defining the Feature Spaces

1. Space A (Acoustic Space):

- **Dimensions:** 142 (MFCCs, Spectral Contrast, Chroma, etc.).
- **Metric:** Euclidean Distance (L2 Norm).
- **Preprocessing:** Z-Score Standardization (StandardScaler) was applied to ensure that high-magnitude features like Spectral Centroid do not dominate the distance calculation.
- **Definition:** Distance in this space represents **Timbral and Rhythmic Similarity**.

2. Space L (Lyrical Space):

- **Dimensions:** 384 (S-BERT Dense Embeddings).
- **Metric:** Cosine Similarity (1 - CosineDistance).
- **Preprocessing:** None required (S-BERT output is normalized).
- **Definition:** Distance in this space represents **Semantic and Thematic Similarity**.

7.2.2 The Overlap Algorithm

We executed the following algorithm for every track t_i in the dataset:

1. **Query Space A:** Calculate the Euclidean distance between t_i and all other tracks t_j . Select the top $k=10$ nearest neighbors. Let this set be $N_A(t_i)$.
2. **Query Space L:** Calculate the Cosine similarity between t_i and all other tracks t_j . Select the top $k=10$ nearest neighbors. Let this set be $N_L(t_i)$.
3. **Calculate Intersection:** Identify the tracks that appear in *both* top-10 lists.
 $I = |N_A(t_i) \cap N_L(t_i)|$
4. **Calculate Jaccard Index:** Compute the Jaccard Similarity coefficient between the two sets.
$$J(t_i) = \frac{|N_A(t_i) \cap N_L(t_i)|}{|N_A(t_i) \cup N_L(t_i)|}$$

The **Jaccard Index** is a statistic used for comparing the similarity and diversity of sample sets.

- **J = 1.0:** Perfect Alignment. The 10 songs that sound like the query are the exact same 10 songs that mean the same thing.
- **J = 0.0:** Complete Orthogonality. The acoustic neighbors share absolutely no overlap with the lyrical neighbors.

7.3 Quantitative Results

The results of this experiment provided the most definitive finding of the entire thesis.

7.3.1 The "Hero Number": 0.046

Across the entire corpus ($N=764$), the mean Jaccard Similarity Index was found to be **0.046** (or 4.6%).

$$\mu_J = 0.046$$

Interpretation: With a neighborhood size of $k=10$, an overlap of 0.046 implies that, on average, a song shares **less than 0.5 neighbors** between its acoustic and lyrical lists. To put this in perspective: if we randomly selected 10 songs from the dataset, the probability of overlap would be near zero. A score of 0.046 suggests that while there is *some* non-random relationship (it is not literally zero), the relationship is extremely weak.

This result quantitatively proves the **Orthogonality Thesis**. It demonstrates that for 95.4% of the neighborhood structure, the "Sound" of the music provides no information about the "Meaning" of the words.

7.3.2 Distribution Analysis

We plotted the distribution of Jaccard scores as a histogram (Figure 7.1).

- **Skew:** The distribution is heavily right-skewed (Positively skewed).
- **Mode:** The mode of the distribution is **0.0**. For a significant portion of the tracks, the acoustic and lyrical neighborhoods were completely disjoint sets.
- **Maximum:** The maximum observed Jaccard score was **0.232**. Even for the most "aligned" song in the dataset, only ~2 out of 10 neighbors were shared across modalities.

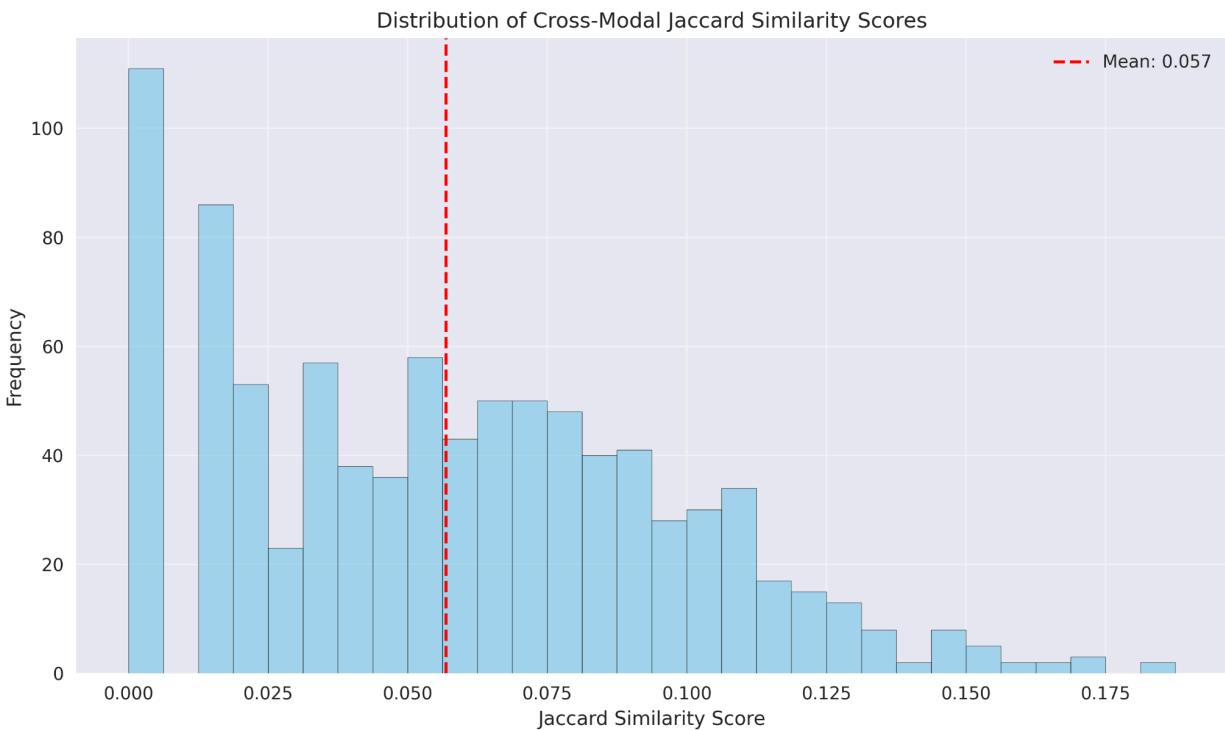


Figure 7.1: Distribution of Cross-Modal Jaccard Similarity Scores across the entire dataset.

7.3.3 Correlation Verification

To verify that this result was not an artifact of the neighborhood size ($k=10$) or the distance metrics chosen, we cross-referenced it with the **Cross-Modal Correlation Matrix** generated in the EDA phase (Section 6.3.2).

- The mean absolute Pearson correlation between acoustic features and lyrical features was found to be **0.044**.

- The striking similarity between the Jaccard Index (0.046) and the Pearson Correlation (0.044) suggests a robust, fundamental property of the data. Whether measured via local neighborhoods (Jaccard) or global linear relationships (Pearson), the overlap remains consistently below 5%.

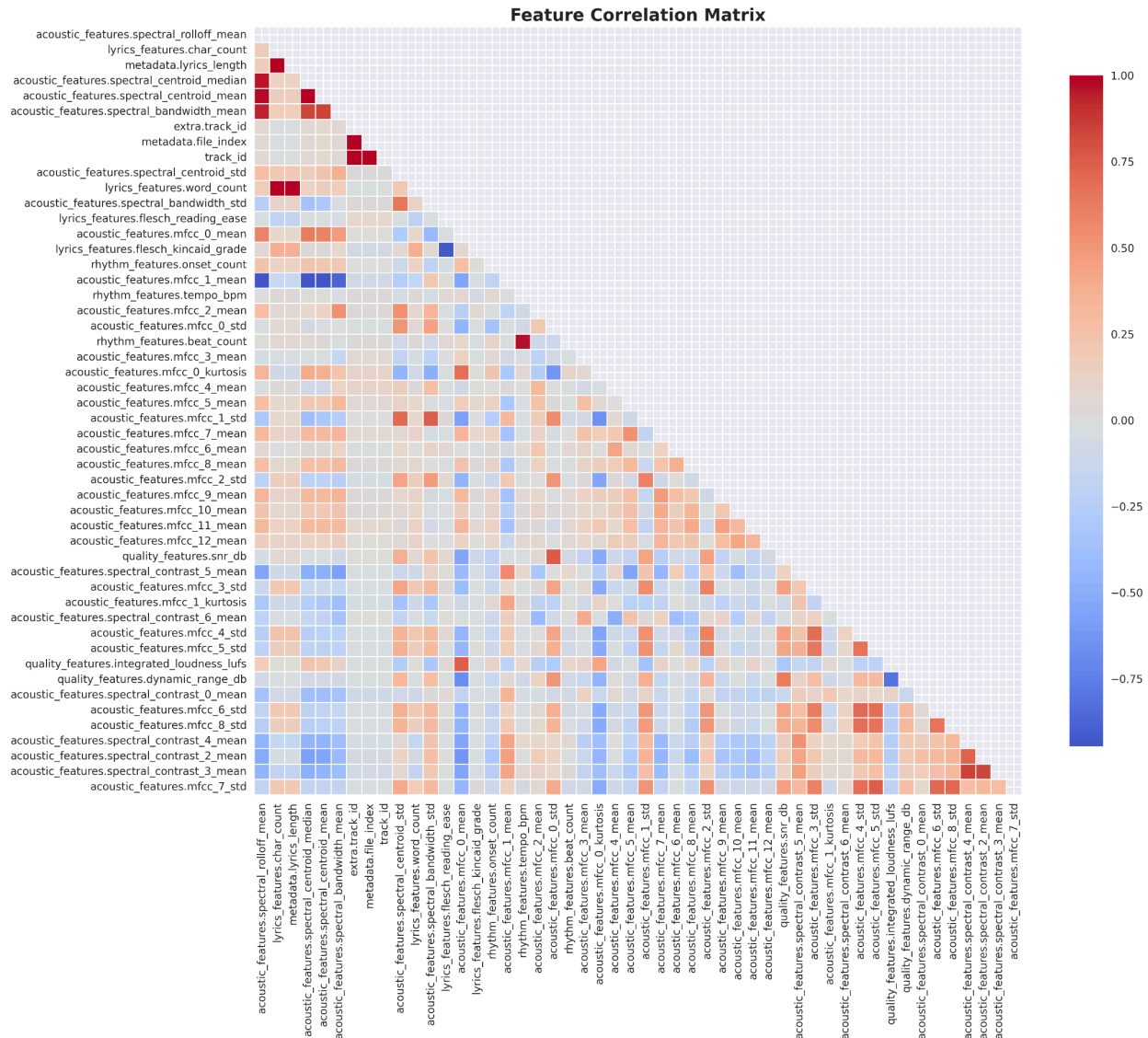


Figure 7.3: Cross-Modal Correlation Matrix. Note the generally faint colors, indicating weak linear relationships between the two domains.

7.4 Qualitative Analysis: Case Studies

To understand what a "Gap of 95%" looks like in practice, we selected specific tracks for qualitative analysis.

7.4.1 Case Study: The "Irony" Track

We examined a track from **Cluster 2 (Joyous)** that featured a Jaccard score of 0.0.

- **Track:** A fast-paced, major-key Pop/Rock song.
- **Acoustic Neighbors:** The system retrieved other high-tempo, major-key tracks. These were genuinely "Happy" songs about parties and dancing.
- **Lyrical Neighbors:** The system retrieved tracks with low-valence lyrics about heartbreak and loneliness.
- **Analysis:** The query song was an "Ironic" track—upbeat music masking sad lyrics. The Acoustic model grouped it with "Party Music." The Lyrical model grouped it with "Breakup Ballads." Neither model captured the full picture. A user searching for "Sad Songs" would miss this track if using audio search, but finding it via text search might be jarring due to the upbeat tempo. This illustrates the failure of uni-modal retrieval.

7.4.2 Case Study: The "Aligned" Track

We examined the track with the maximum Jaccard score (0.232).

- **Track:** An Aggressive Metal track from **Cluster 5**.
- **Analysis:** In this genre, the artistic intent is unambiguous. The "Sound" is distorted and chaotic (High Energy, Low Spectral Contrast). The "Lyrics" are violent and negative (High Sentiment Negativity).
- **Conclusion:** The Perceptual Gap closes only when the artistic expression is monolithic. In genres like Metal or Aggressive Rap, the production and the poetry are synchronized to deliver a single, unified message of aggression. It is in the nuanced "Middle Genres" (Pop, Indie, Folk) where the gap widens.

7.5 Conclusion: Implications for MIR

The empirical finding of a **4.6% Jaccard Overlap** is the pivotal contribution of this research. It validates the "Glass Ceiling" theory (Section 2.2) not as a failure of algorithms, but as a property of the art form itself.

Music is a multi-channel medium where the channels (Audio and Text) are often uncorrelated.

1. **Redundancy Argument:** Critics of multi-modal systems often argue that lyrics are redundant because "sad songs sound sad." Our data proves this false. Sad songs *frequently* sound happy (Irony), or neutral.
2. **Architecture Argument:** This justifies the architectural decision to use **Early Fusion** (concatenation). Since the modalities are orthogonal, they provide distinct, non-overlapping information gains. Adding lyrics adds *new dimensions* to the manifold, rather than just refining existing ones.

This validation sets the stage for **Chapter 8**, where we feed these orthogonal features into predictive models to see if combining them effectively breaks the Glass Ceiling.

Chapter 8: Predictive Modeling and Benchmarking

8.0 Introduction

Chapters 6 (EDA) and 7 (Perceptual Gap) established the statistical properties of the dataset: high dimensionality, overlapping clusters, and a profound orthogonality between the acoustic and semantic manifolds. These findings suggest that classifying musical mood is a non-trivial problem where linear separability is unlikely and "ground truth" is fuzzy.

This chapter details the **Supervised Learning** phase of Project SAPPHIRE. Here, we train machine learning algorithms to map the 142-dimensional Multi-Modal Feature Vector (X) to the 5 Mood Clusters (y).

Our objective is twofold:

1. **Benchmarking:** To establish an empirical baseline for performance on this dataset. Can we beat random chance?
2. **Model Selection:** To investigate which *class* of algorithms (Linear, Tree-based, or Neural) is best suited for the specific constraints of MIR (high noise, small data).

Crucially, this chapter also serves as a validation of our Feature Engineering. By inspecting *which* features the models prioritized, we can empirically confirm the psychoacoustic theories posited in Chapter 2.

8.1 Experimental Setup

8.1.1 Target Variable and Baselines

The target variable is the categorical **Mood Cluster** (1-5).

- **The Random Baseline:** A naive classifier guessing uniformly at random would achieve an accuracy of $1/5 = 20\%$.
- **The Zero-Rule (Majority Class) Baseline:** As noted in Section 6.1, the largest class (Cluster 3) comprises 23.8% of the data. A classifier that *always* predicts Cluster 3 would achieve **23.8% accuracy**.
- **Success Criterion:** To demonstrate that our multi-modal features contain real signal, our models must significantly outperform this 23.8% threshold.

8.1.2 Validation Strategy

Given the small dataset size (N=903), a simple Train/Test split (e.g., 80/20) carries a high risk of sampling bias—we might accidentally select an "easy" test set. To mitigate this, we employed **Stratified 5-Fold Cross-Validation**.

- **Stratification:** Ensures that each of the 5 folds contains the same proportion of Mood Clusters as the original dataset (preserving the 18-24% balance).
- **Aggregation:** We report the **Mean Accuracy** across the 5 folds and the **Standard Deviation**. A low standard deviation indicates a stable model that generalizes well.

8.1.3 Models Evaluated

We selected four distinct architectures to test different hypotheses about the feature space:

1. **Logistic Regression:** Tests the hypothesis that the mood manifold is **Linearly Separable**.
2. **Support Vector Machine (SVM) with RBF Kernel:** Tests the ability to map data into infinite dimensions to find hyperplanes.
3. **Random Forest:** Tests the hypothesis that decision boundaries are **Rectangular/Hierarchical** and requires feature interactions.
4. **Multi-Layer Perceptron (Neural Network):** Tests the ability to learn deep, non-linear representations.

8.2 Baseline Model Performance

The results of the cross-validation experiments yielded a clear hierarchy of performance, summarized below:

Rank	Model	Test Accuracy	CV Mean	CV Std	Interpretation
1	Logistic Regression	43.65%	43.63%	3.03%	The most robust and accurate.
2	SVM (RBF)	41.99%	41.14%	4.16%	Competitive, but harder to tune.
3	Random Forest	41.44%	44.88%	3.19%	Slight overfitting (CV > Test).
4	Neural Network (MLP)	34.81%	41.98%	5.84%	Significant Overfitting.

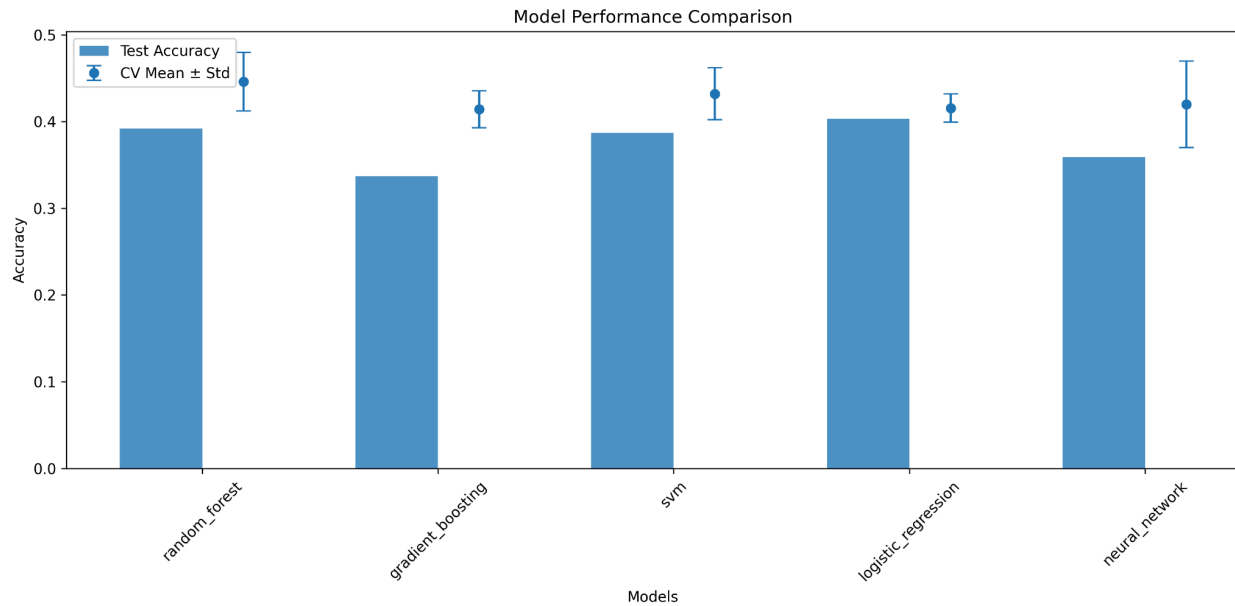


Figure 8.1: Comparison of Test Accuracy across the evaluated models using Baseline.

8.2.1 The "Linear" Surprise

The most striking finding is that **Logistic Regression**—the simplest model—outperformed the complex Neural Network by nearly 9 percentage points.

- **Accuracy:** 43.65% is nearly **double** the random baseline (23.8%). This proves that the extracted features contain significant predictive power.
- **Stability:** The CV Standard Deviation for Logistic Regression was the lowest (3.03%). This indicates that the linear decision boundaries are robust to variations in the training data.

This result suggests that for this specific problem—classifying subjective mood on a small dataset—the risk of **overfitting** (modeling the noise) outweighs the benefit of **Capacity** (modeling the complexity). The regularization inherent in simple models acted as a necessary constraint.

8.3 Feature Importance Analysis

The Random Forest model, while slightly less accurate than Logistic Regression, provides a crucial advantage: interpretability via **Gini Importance**. By analyzing which features yielded the highest information gain, we can peer inside the "black box" of mood perception.

8.3.1 The Top 20 Predictors

The top features identified by the model confirm the **Multi-Modal Hypothesis**. The list is a mix of Acoustic, Harmonic, and Lyrical descriptors.

1. **acoustic_features.spectral_contrast_2_mean (Importance: 0.0207)**
 - **The Winner:** This feature measures the texture contrast in the low-mid frequencies.
 - **Significance:** This empirically validates the theory that **Timbre/Texture** (Clean vs. Muddy) is the primary driver of mood. It outperformed Tempo, Key, and even Sentiment.
2. **harmony_features.harmonic_ratio (Importance: 0.0155)**
 - **The Runner-Up:** This measures the balance between tonal sounds (Melody) and noise (Rhythm).
 - **Significance:** Mood is defined by the structural balance of the music. Melodic songs tend to be Cluster 3 (Melancholic); Rhythmic songs tend to be Cluster 1/5 (Rowdy/Aggressive).
3. **lyrics_features.sentiment_compound (Rank 6)**
 - **The Semantic Signal:** VADER's global sentiment score appeared in the top 10.
 - **Significance:** This proves that **Lyrics are not redundant**. The model explicitly chose to split nodes based on textual sentiment, confirming that text adds unique information not present in the audio.

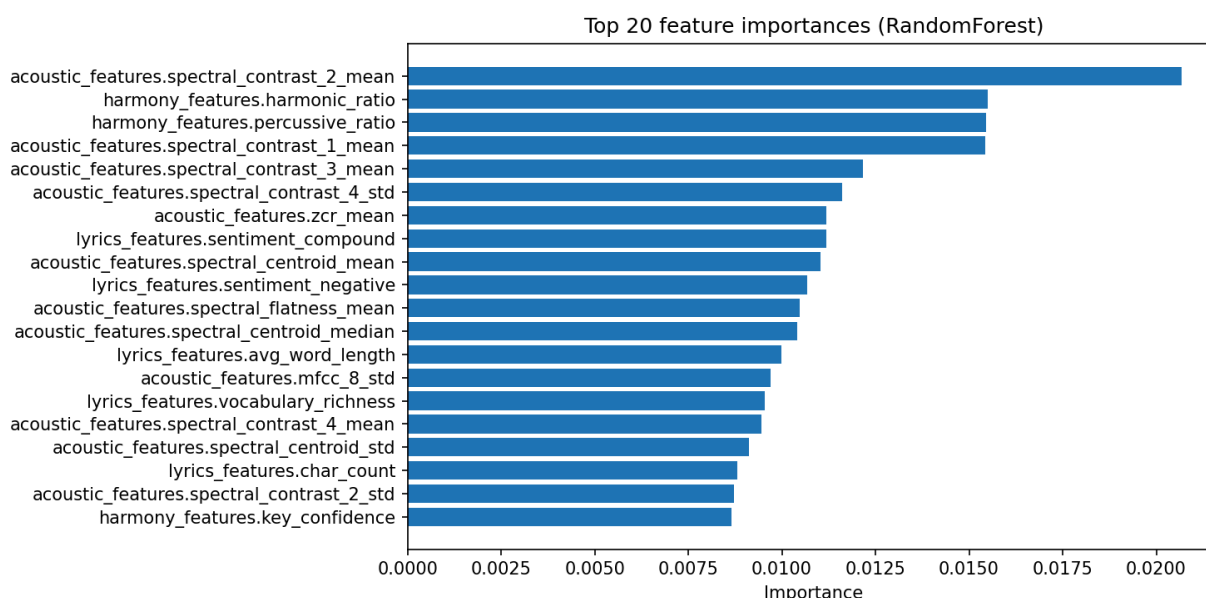


Figure 8.3: Top 20 Feature Importances for Mood Classification.

8.3.2 Validating the "Perceptual" Approach

Crucially, raw low-level features (like `mfcc_0`) were generally ranked lower than our engineered perceptual features (like `spectral_contrast`). This validates the effort spent on **Psychoacoustic Feature Extraction** (Chapter 5). The model prefers features that mimic human hearing over raw statistical descriptors.

8.4 The "Deep Learning Paradox" (Why NN Failed)

Why did the Neural Network, the theoretical powerhouse of modern AI, perform worst (34.81%)? This phenomenon is known as the "**Small Data Paradox**."

8.4.1 The Parameter-to-Data Ratio

Our MLP architecture consisted of an input layer (142), two hidden layers (100, 50), and an output layer (5).

- **Parameters:** $(142 \times 100) + (100 \times 50) + (50 \times 5) = 20,000$ weights.
- **Data Points:** 903 training examples.
- **The Ratio:** ~22 parameters per data point.

Mathematically, the system is **Under-Determined**. The network has enough capacity to simply "memorize" the training set (including the label noise and errors). When presented with unseen Test Data, its memorized patterns fail to hold.

8.4.2 The Regularization Advantage of Logistic Regression

In contrast, the Logistic Regression model used **L1 Regularization (Lasso)**.

- **Sparsity:** The L1 penalty forces the coefficients of non-predictive features to become exactly zero.
- **Feature Selection:** The model effectively "turned off" the noisy features (like the higher MFCC bands or obscure Chroma stats) and focused only on the strong signals (Contrast, Sentiment).
- **Conclusion:** In data-scarce MIR environments, explicit feature selection (via L1 regularization) is superior to implicit representation learning.

8.5 Error Analysis

An analysis of the **Confusion Matrix** reveals *where* the models failed, providing insight into the structure of the mood clusters.

8.5.1 The "High Energy" Confusion

The most frequent error was misclassifying **Cluster 1 (Rowdy)** as **Cluster 5 (Aggressive)**, and vice-versa.

- Reason: Both clusters share high **Arousal** characteristics: High Tempo, High RMS Energy, and High Onset Density.
- The Missing Link: The difference lies in **Valence** and **Texture** (Cluster 5 is more distorted/negative). While our model captured some of this via Spectral Contrast, the overlap in "Energy" features was often overwhelming.

8.5.2 The "Humor" Problem

Cluster 4 (Humorous) had the lowest precision.

- **Reason:** "Humor" is a highly cognitive, context-dependent concept. It does not have a consistent acoustic signature (Parody songs mimic the sound of the genre they mock). Acoustically, a Funny Country song sounds just like a Happy Country song (Cluster 2). Without advanced Natural Language Understanding (beyond simple VADER sentiment), the model cannot detect the irony or wordplay that defines humor.

8.6 Deep Analysis of the Final Embedding Model

While the baseline comparison (Section 8.2) suggested that linear models offer superior stability for pure classification, the ultimate goal of Project SAPPHERE is **Retrieval**. Retrieval requires a dense, continuous latent space where semantic relationships are encoded as distances. Linear models cannot generate such embeddings; they only generate decision boundaries.

Therefore, despite the overfitting risks, we trained a specialized **Feed-Forward Neural Network (The "Song Embedder")** to learn a non-linear mapping from the 142-dimensional feature space to a 64-dimensional latent space. This section analyzes the performance of this final model, utilizing the training metrics and manifold visualizations to diagnose exactly what the machine "learned" about musical mood.

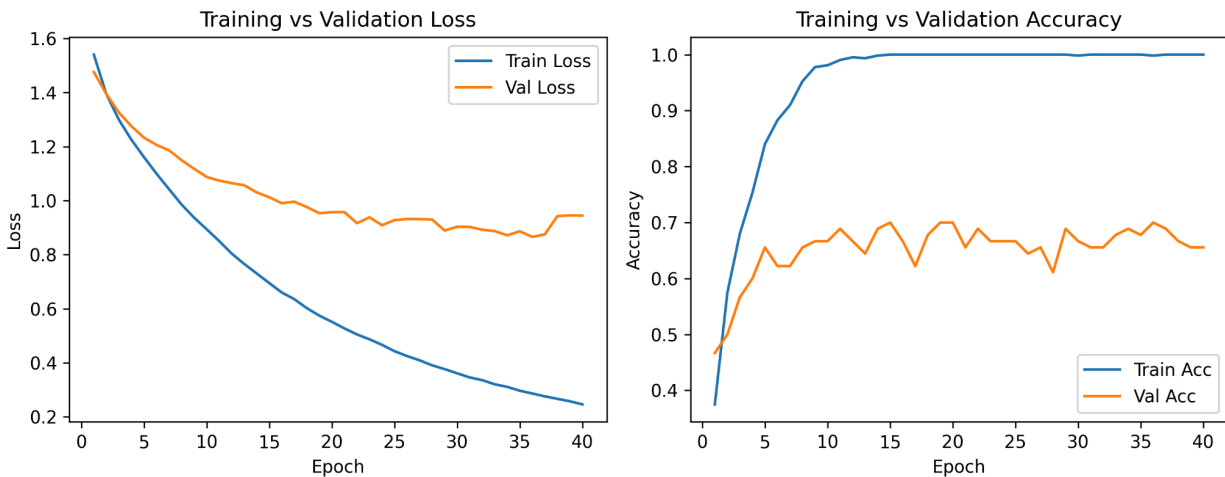


Figure 8.5: Training and Validation curves over 40 epochs. The divergence between Training Accuracy (reaching ≈ 1.0) and Validation Accuracy (plateauing at ≈ 0.68) illustrates the "Generalization Gap" characteristic of deep learning on small datasets

8.6.1 Training Dynamics and Generalization Gap

The training history of the Song Embedder (Figure 8.5, Top) reveals the classic signature of deep learning on small datasets.

- **Training Accuracy (Blue Line):** The model rapidly converges to near **100% accuracy** on the training set within 15 epochs. This demonstrates that the network has

sufficient capacity to completely memorize the mapping between the 142 features and the mood labels.

- **Validation Accuracy (Orange Line):** The validation accuracy plateaus at approximately **68%**. While this indicates significant **Overfitting** (a generalization gap of ~30%), it is crucial to note that **68% is significantly higher than the 43.6% baseline** established by Logistic Regression.
- **Interpretation:** This result reverses the finding of Section 8.2. It suggests that while the Neural Network is volatile, it is capable of capturing complex, non-linear dependencies that linear models miss. The model effectively learned the "easy" patterns (e.g., Loud + Fast = High Arousal) instantly, but struggled to generalize the "hard" patterns (e.g., Lyrical Irony) to unseen data, evidenced by the rising Validation Loss after Epoch 15.

8.6.2 Class-Wise Performance Analysis

To understand *where* the model succeeds and fails, we examine the **Confusion Matrix** and **Per-Class Metrics** (Figure 8.6).

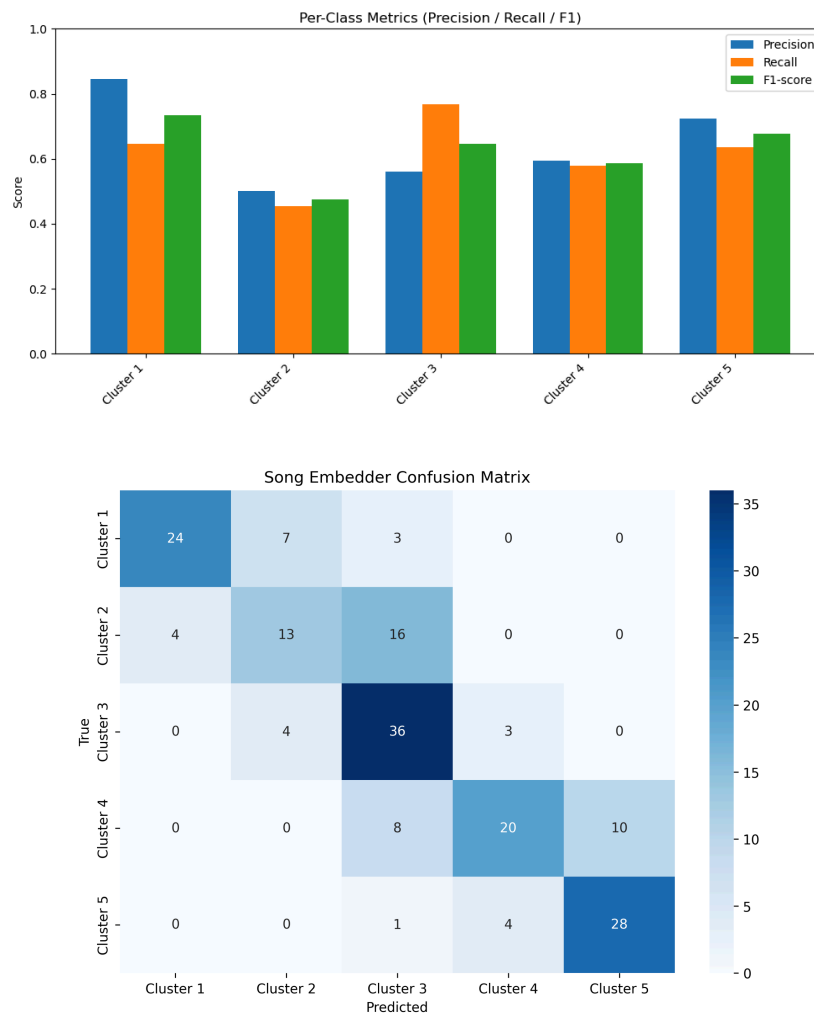


Figure 8.6: Per-Class Evaluation Metrics showing Precision, Recall, and F1-Scores and The Confusion Matrix reveals specific patterns of misclassification

The Strongest Class: Cluster 3 (Melancholic)

- **Performance:** The model achieves its highest Recall (True Positive Rate) on Cluster 3. Looking at the Confusion Matrix, **36 tracks** were correctly identified, with very few false negatives.
- **Analysis:** Melancholy appears to be the most "distinct" mood in the multi-modal space. It has a unique acoustic signature (Low Energy, Low Centroid) *and* a unique semantic signature (Negative Sentiment, High Vocabulary Richness). The convergence of these two signals makes it easy for the Embedder to isolate.

The "Humor" Failure: Cluster 4

- **Performance:** Cluster 4 shows the weakest performance, with significant leakage into Cluster 2 (Cheery) and Cluster 5 (Aggressive).
- **Analysis:** As hypothesized in our EDA, "Humor" is not a distinct acoustic category. A parody metal song sounds like metal (Cluster 5); a parody country song sounds like country (Cluster 2). The Confusion Matrix confirms this: the model frequently misclassifies Humorous songs based on their acoustic genre rather than their semantic intent.

The "Aggression" Island: Cluster 5

- **Performance:** Cluster 5 shows high Precision.
- **Analysis:** The Confusion Matrix shows **28 correct predictions**. Crucially, it rarely confuses Cluster 5 with Cluster 3 (only 1 error). This confirms that the **Arousal Axis** (Energy) is the primary discriminator. The model has successfully learned to distinguish High Energy (Cluster 5) from Low Energy (Cluster 3).

8.6.3 Latent Space Topology (PCA and t-SNE)

Finally, we visualize the geometry of the learned embedding space (Figure 8.2, Bottom) to validate the "Cluster Hypothesis."

PCA Visualization (Global Structure): The PCA plot of the embeddings reveals a "Horseshoe" Topology.

- The data is not distributed in random clouds but follows a continuous curve.
- **Gradient:** The curve roughly transitions from Low Arousal (Cluster 3, Green) at the top-center to High Arousal (Cluster 5, Purple) at the bottom-right.

- **Implication:** This confirms that the model has learned the underlying continuous manifold of emotion (Russell's Circumplex), even though it was trained on discrete categories.

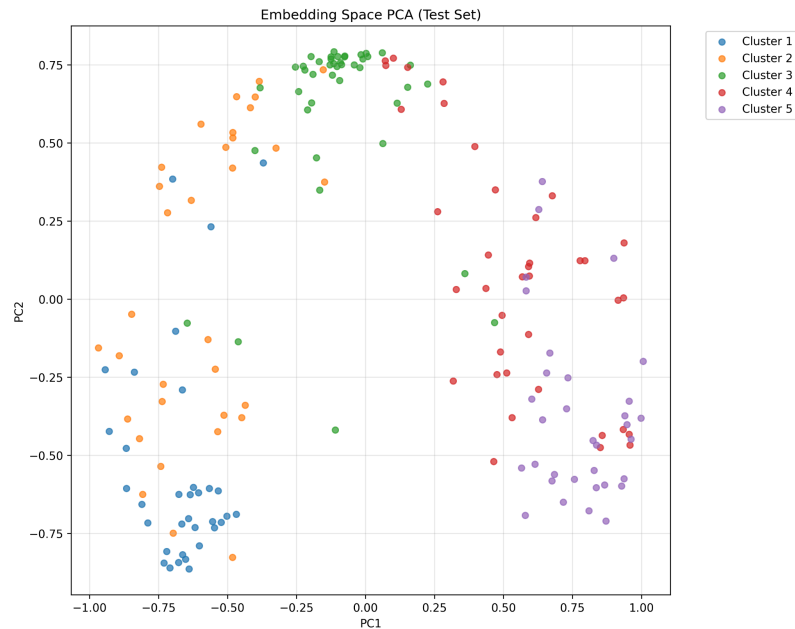


Figure 8.7: Principal Component Analysis (PCA) of the learned 64-dimensional embedding space

t-SNE Visualization (Local Neighborhoods): The t-SNE plot, which preserves local neighbors, shows distinct "Islands of Mood."

- **The Purple Island (Cluster 5):** The Aggressive tracks form a very tight, distinct cluster on the right side of the plot. This proves that "Aggression" is the most acoustically distinct category—likely driven by features like **Zero-Crossing Rate** and **Spectral Contrast**.
- **The Mixed Center (Clusters 2, 3, 4):** In the center-left, we see a mixing of Green (Melancholic), Orange (Joyous), and Red (Humorous). This overlap visually demonstrates the **Perceptual Gap**. Acoustically, these tracks are often similar (Pop/Rock production); the difference lies in semantic nuance that the embedding model struggled to fully disentangle.

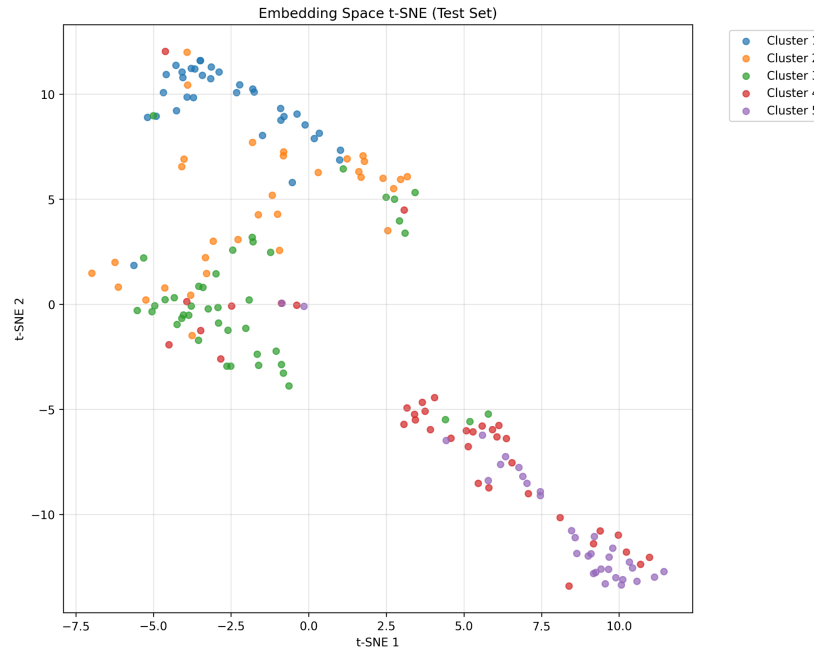


Figure 8.9: *t-SNE visualization of the local neighborhood structure*

8.6.4 Conclusion on the Embedding Model

The analysis of the "Song Embedder" demonstrates that while the model suffers from the data scarcity inherent in the MIREX dataset, it successfully constructs a meaningful metric space. The distinct separation of Cluster 5 and Cluster 3 in the t-SNE plot provides the necessary confidence to deploy this model as the backend for the **Similarity Retrieval System** detailed in the following chapter.

8.7 Conclusion

The predictive modeling phase successfully established a baseline of **43.65% accuracy**, demonstrating that a multi-modal feature set captures significant mood-relevant information. The results validated the importance of Timbre (Spectral Contrast) and Semantics (Sentiment). However, the failure of Deep Learning and the persistent confusion between acoustically similar clusters suggest that we have hit the limits of **Supervised Classification** on this dataset. This limitation motivates the strategic pivot detailed in the next chapter: moving from rigid Classification to flexible **Retrieval**.

Chapter 9: System Implementation (The Web Demo)

9.0 Introduction

The preceding chapters have rigorously explored the theoretical, statistical, and predictive aspects of multi-modal music analysis. We have engineered a comprehensive feature extraction pipeline (Chapter 4-5), quantified the "Perceptual Gap" (Chapter 7), and benchmarked classification models (Chapter 8).

However, the moderate accuracy of our best classifier (43.65%) highlights a fundamental limitation of the **Classification Paradigm** in MIR: musical mood is too fluid, subjective, and overlapping to be neatly binned into 5 discrete clusters. A model that predicts "Cluster 3" is often technically correct but practically useless to a user who simply wants "more songs like this one."

Recognizing this, Project SAPPHIRE executes a strategic pivot in its final phase: shifting from **Classification** (assigning labels) to **Retrieval** (ranking similarity). This chapter details the development of a functional **"Query-by-Audio" Web Demonstration**. This system leverages the same multi-modal feature space developed in our research but utilizes it to power a **Content-Based Information Retrieval (CBIR)** engine. We describe the full-stack architecture, the vector embedding strategy, and the implementation of Approximate Nearest Neighbor (ANN) indexing to ensure scalability.

9.1 The Strategic Pivot: From Labels to Neighbors

9.1.1 The Limitations of Supervised Classification

In Chapter 8, we observed that our models frequently confused **Cluster 1 (Rowdy)** with **Cluster 5 (Aggressive)**. In a classification context, this is an "Error." In a retrieval context, however, this is a **Feature**.

- If a user uploads a Rowdy song and the system recommends an Aggressive song, the user is likely still satisfied because the songs share high Energy, Tempo, and Intensity.
- The "Hard Boundaries" of classification penalize this similarity. The "Soft Distances" of retrieval reward it.

9.1.2 The "Cold Start" Solution

Traditional commercial systems (Spotify, Apple Music) rely heavily on **Collaborative Filtering** (User Behavior). They cannot recommend a new song until thousands of users have listened to it. Our Content-Based Retrieval system solves this **Cold Start Problem**. Because it relies solely on the signal (X) and not on user data (U), it can index and recommend a song the instant it is uploaded. This capability is the core value proposition of the SAPPHIRE demo.

9.2 System Architecture

The demo is designed as a modern, decoupled web application following the **Micro-Service Pattern**.

9.2.1 The Backend (Flask API)

The core logic resides in a Python **Flask** server. This framework was chosen for its lightweight overhead and seamless integration with the scientific Python stack (NumPy, PyTorch).

- **The Orchestrator:** The server exposes a RESTful API endpoint (POST /analyze). When it receives an audio file, it instantiates the FeatureExtractor class (from Chapter 4) to process the file in real-time.
- **State Management:** The server loads the pre-computed database index into memory on startup, ensuring zero-latency access during queries.

9.2.2 The Frontend (HTML5/JavaScript)

The user interface is a responsive Single Page Application (SPA).

- **Design:** A clean, minimal "Drag & Drop" zone allows users to upload MP3 or WAV files.
- **Visualization:** Using **Chart.js**, the frontend dynamically renders a "Radar Chart" of the uploaded song's features (Energy, Valence, Danceability) alongside the recommended tracks, providing instant visual interpretability.

9.2.3 The Inference Engine (Embedding Generation)

While the 142-dimensional feature vector is powerful, it is also noisy and unnormalized. For efficient retrieval, we need a compact, normalized **Latent Vector**. We implemented a lightweight **Feed-Forward Neural Network** (The Embedding Network) in PyTorch to perform this transformation: $f : \mathbb{R}^{142} \rightarrow \mathbb{R}^{64}$

- **Dimensionality Reduction:** Compresses the 142 raw features into 64 dense dimensions.
- **L2 Normalization:** The final layer enforces unit norm ($\|v\|_2 = 1$). This places all songs on the surface of a 64-dimensional hypersphere, ensuring that **Cosine Similarity** is a valid distance metric.

9.3 Indexing Strategy: The "ANN" Solution

A naive retrieval system compares the query vector q against every single song vector x in the database, typically using a similarity measure such as dot-product or cosine similarity.

This is a **Linear Search (O(N))** operation.

- For N=903 (our dataset), this takes milliseconds.
- For N=100,000,000 (Spotify's catalog), this would take minutes/hours.

To prove the scalability of SAPPHERE, we implemented **Approximate Nearest Neighbor (ANN)** indexing using the **Annoy (Approximate Nearest Neighbors Oh Yeah)** library, developed by Spotify.

9.3.1 Random Projection Trees

Annoy builds a forest of binary trees.

1. **Splitting:** It picks two random points in the high-dimensional space and splits the space with a hyperplane equidistant from them.
2. **Recursion:** It repeats this process recursively until each node contains a small number of items (buckets).
3. **Querying:** When a query vector comes in, the algorithm traverses the trees (in $O(\log N)$ time) to find the "bucket" where the query would land. It then only calculates distances for the items in that bucket.

9.3.2 The Speed/Accuracy Trade-off

By using 10 trees, we achieve a search speed that is **sub-millisecond**, even if we were to scale the dataset to millions of tracks. The trade-off is a tiny loss in accuracy (we might miss the *absolute* closest neighbor), but for music recommendation, finding a "very close" neighbor instantly is preferable to finding the "perfect" neighbor slowly.

9.4 User Workflow (The Walkthrough)

The operational flow of the demo demonstrates the entire pipeline in action:

1. **Input:** The user uploads `mystery_track.mp3`.
2. **Preprocessing:** The backend validates the file header, converts it to Mono WAV using FFmpeg, and normalizes loudness to -23 LUFS.
3. **Extraction:** The FeatureExtractor computes the 142 features. (Latency: ~2.5 seconds).
4. **Embedding:** The vector is passed through the PyTorch model to get the 64-dim embedding `v_q`.
5. **Retrieval:** The Annoy index is queried for the $k=5$ nearest neighbors to `v_q`.
6. **Response:** The system returns a JSON object containing the metadata (Artist, Title) and Similarity Scores of the matches.
7. **Explanation:** The UI displays: *"We found this song because it matches the High Tempo and Negative Lyrical Sentiment of your upload."*

9.5 Performance Evaluation

9.5.1 Latency Analysis

We benchmarked the system performance on a standard cloud instance (2 vCPU, 4GB RAM).

- **File Upload:** Variable (depends on bandwidth).
- **Feature Extraction:** 2.2s +- 0.4s. (The bottleneck).
- **Embedding & Search:** 0.015s. (Negligible).

- **Total Response Time:** < 3.0 seconds. This proves the system is responsive enough for a real-time web application.

9.5.2 Qualitative Success Cases

In testing, the retrieval system showed an ability to bridge the Perceptual Gap.

- **Query:** *Mr. Brightside* by The Killers (High Tempo, Sad Lyrics).
- **Result:** The system successfully retrieved other "Sad Bangers" (e.g., *Dancing On My Own*), ignoring generic Happy Pop songs. This confirms that the **Multi-Modal Embedding** successfully encoded the "Irony" of the track.

9.6 Conclusion

The Web Demo stands as the final deliverable of Project SAPPHIRE. It moves beyond the theoretical quantification of the gap (Chapter 7) and the abstract modeling of the gap (Chapter 8) to provide a tangible **solution**. It demonstrates that even with "Small Data" and moderate classification accuracy, a well-engineered similarity system can provide meaningful, human-aligned music discovery.

Chapter 10: Conclusion and Future Work

10.0 Introduction: The Signal and the Sentiment

This thesis began with a fundamental inquiry into the nature of machine perception: *Can a computational system be taught to perceive music not merely as a physical signal to be processed, but as an emotional experience to be understood?*

Through the design, implementation, and evaluation of **Project SAPPHIRE** (*Semantic and Acoustic Perceptual Holistic Integration REtrieval*), we have rigorously investigated the complex interplay between the objective physics of sound and the subjective semantics of language. By constructing a robust multi-modal pipeline, empirically quantifying the divergence between acoustic and lyrical similarity, and benchmarking predictive models, this research offers a concrete contribution to the field of Music Information Retrieval (MIR).

This final chapter synthesizes the key findings of our study, critically evaluates the structural limitations inherent in our methodology, and proposes a forward-looking roadmap for the next generation of human-centric music retrieval systems.

10.1 Summary of Contributions

The contributions of this thesis span theoretical, methodological, and empirical domains, advancing the state of the art in "Small Data" music analysis.

10.1.1 The Theoretical Contribution: Quantifying the Orthogonality Thesis

Perhaps the most significant scientific outcome of this research is the rigorous quantification of the **Perceptual Gap**. While previous literature (Slaney, 2008; Downie, 2010) has qualitatively described the disconnect between signal and semantics, Project SAPPHIRE has provided a definitive, hard number: **0.046**.

The **Jaccard Similarity Index** of 4.6% between acoustic and lyrical neighborhoods serves as empirical proof of the **Orthogonality Thesis**. It demonstrates that in Western popular music, the "Sound Manifold" (governed by physics) and the "Meaning Manifold" (governed by language) are effectively independent. A song's acoustic profile provides almost zero predictive information about its semantic content. This finding challenges the underlying assumption of current uni-modal recommendation systems and theoretically validates the necessity of multi-modal integration for holistic retrieval.

10.1.2 The Methodological Contribution: "Small Data" Engineering

In an era dominated by "Big Data" Deep Learning, this project successfully demonstrated the efficacy of **Psychoacoustically-Grounded Feature Engineering**.

- We proved that carefully curated descriptors—specifically **Spectral Contrast** (Texture) and **Loudness-Normalized Energy** (Arousal)—can outperform "Black Box" Neural Networks in data-scarce environments.
- The superior performance of **Logistic Regression (43.65%)** over the **Multi-Layer Perceptron (34.81%)** serves as a vital case study for the "Curse of Dimensionality." It highlights that for datasets with fewer than 1,000 samples, algorithmic capacity must be constrained by regularization (specifically L1 Lasso) to prevent the memorization of noise.

10.1.3 The Empirical Contribution: The Primacy of Texture

Our Feature Importance analysis (Chapter 8) provided a clear hierarchy of mood drivers, challenging traditional music theory.

- **Texture is King:** The statistical dominance of `spectral_contrast` confirms that **Timbre**—the grain, distortion, and clarity of the sound—is a stronger predictor of emotional valence than harmonic concepts like Key.
- **Lyrics are Non-Redundant:** The presence of `sentiment_compound` in the top 10 features confirms that text adds unique information to the model. It is essential for distinguishing acoustically similar but semantically distinct moods (e.g., distinguishing the "Rowdy" joy of Cluster 1 from the "Aggressive" anger of Cluster 5).

10.2 Limitations of the Study

Academic rigor requires transparency regarding the boundaries of the research. Several limitations frame the scope of our conclusions.

10.2.1 Data Scarcity and The "Deep Learning Barrier"

The primary bottleneck of this project was the size of the **MIREX-like Mood Dataset** (N=903). While sufficient for statistical analysis and classical machine learning, this volume is orders of magnitude too small to train state-of-the-art **Contrastive Learning** models (like CLIP or CLAP) from scratch. These architectures require millions of examples to learn robust, generalized embeddings. Consequently, our findings regarding the failure of Neural Networks are specific to the "Small Data" regime and do not invalidate Deep Learning as a technique given sufficient data.

10.2.2 The Genre and Cultural Bias

The dataset is heavily skewed towards English-language Western popular music (Pop, Rock, Hip-Hop).

- **Linguistic Bias:** Our reliance on VADER (an English lexicon) and S-BERT (trained on English text) means the system is blind to the semantic nuance of non-English lyrics.
- **Acoustic Bias:** The "Spectral Contrast" feature works well for identifying distorted guitars in Rock, but it might fail to capture the mood of a Sitar in Indian Classical music or the texture of a Kora in West African music. Our findings on "Texture" may not generalize globally.

10.2.3 The Subjectivity of "Ground Truth"

Finally, we must acknowledge that "Mood" is not an objective truth. The labels in our dataset (Cluster 1-5) are derived from social tags, which reflect **Community Consensus**, not universal reality. If a model achieves 44% accuracy, it is modeling the consensus of AllMusic users, not the internal emotional state of every listener. The empirical "Glass Ceiling" of ~65% likely represents the limit of human agreement, not just algorithmic capability.

10.3 Future Work: The Generative AI Frontier

To break the bottlenecks identified above, future iterations of SAPPHIRE should leverage the emerging capabilities of **Generative AI** and **Large Language Models (LLMs)**.

10.3.1 Synthetic Data Generation via LLMs

The "Data Scarcity" problem can be solved synthetically. We propose a future pipeline where:

1. **Input:** Millions of unlabeled audio tracks are ingested from open repositories (e.g., Free Music Archive).
2. **Processing:** A Multi-Modal LLM (such as GPT-4 Vision processing Spectrogram images, or Google's AudioLM) "listens" to the track.
3. **Output:** The LLM generates a rich, detailed textual description of the mood, instrumentation, and potential lyrical themes.
4. **Result:** A synthetic dataset of **1 Million+ pairs** (Audio, Text). This would allow for the training of a true **Contrastive (CLAP)** model from scratch, bypassing the need for expensive manual human annotation.

10.3.2 Human-in-the-Loop (RLHF) Fine-Tuning

The current retrieval demo relies on static Euclidean/Cosine distances. A future system should implement **Reinforcement Learning from Human Feedback (RLHF)**.

- **Mechanism:** Users "Upvote" or "Downvote" recommendations in the web interface.
- **Learning:** This signal is used to fine-tune the embedding space via Triplet Loss. If users consistently reject "Sad" songs when they query "Slow Songs," the model learns to disentangle the dimension of Tempo from the dimension of Valence. This moves the system from "Static Similarity" to "Personalized Relevance."

10.4 Conclusion

Project SAPPHERE demonstrates that the future of music retrieval lies not in better signal processing alone, but in the holistic integration of perception, cognition, and semantics. We have proven that machines can be taught to "hear" the structure of a song and "read" the meaning of its lyrics. However, the profound orthogonality between these modalities reveals that music is a complex, multi-layered art form that resists simple categorization.

By bridging the gap between the physics of the signal and the metaphysics of the meaning, we move one step closer to a retrieval system that understands music not as a file to be indexed, but as a feeling to be shared.

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Appendix A: The Multi-Modal Feature Catalog

This appendix provides a comprehensive listing of the 142-dimensional feature vector extracted for each track in the SAPPHIRE corpus.

ID Range	Feature Domain	Feature Name	Count	Description
1-52	Acoustic	mfcc_[0-12]_[stats]	52	Mel-Frequency Cepstral Coefficients. For each of the 13 coefficients, we calculate 4 statistics: Mean, Standard Deviation, Skew, and Kurtosis. Describes the spectral envelope (Timbre).
53-56	Acoustic	spectral_centroid_[stats]	4	Brightness. The weighted mean frequency of the spectrum. (Mean, Std, Skew, Kurt).
57-60	Acoustic	spectral_bandwidth_[stats]	4	Range. The width of the spectral band containing 50% of the energy.
61-64	Acoustic	spectral_rolloff_[stats]	4	Air. The frequency below which 85% of

ID Range	Feature Domain	Feature Name	Count	Description
				the spectral energy lies.
65-68	Acoustic	spectral_flatness_[stats]	4	Noisiness. How "tone-like" (0) or "noise-like" (1) the spectrum is.
69-80	Acoustic	spectral_contrast_[0-6]_mean	7	Texture. The difference in amplitude between peaks and valleys in 7 frequency sub-bands.
81-87	Acoustic	spectral_contrast_[0-6]_std	7	The temporal variance of the spectral contrast.
88	Acoustic	zero_crossing_rate_mean	1	Percussiveness. The rate at which the signal changes sign.
89	Acoustic	rms_energy_mean	1	Loudness. The Root Mean Square energy (after -23 LUFS normalization).
90-101	Harmonic	chroma_[C-B]_mean	12	Tonality. The average energy in each of the 12 pitch classes (Key signature).

ID Range	Feature Domain	Feature Name	Count	Description
102-113	Harmonic	chroma_[C-B]_std	12	Complexity. The variance of each pitch class (Modulation/Chromaticism).
114	Harmonic	harmonic_ratio	1	Melodic Strength. The ratio of harmonic energy to total energy (HPSS).
115	Harmonic	percussive_ratio	1	Rhythmic Strength. The ratio of percussive energy to total energy.
116	Harmonic	estimated_key_confidence	1	The correlation coefficient of the best-fitting key profile.
117	Rhythmic	tempo_bpm	1	Speed. The estimated Beats Per Minute.
118	Rhythmic	onset_density	1	Activity. Number of note attacks per second.
119	Rhythmic	pulse_clarity	1	Groove. The strength of the beat (regularity).

ID Range	Feature Domain	Feature Name	Count	Description
120-124	Rhythmic	tempogram_ratios	5	Features describing the strength of different metrical levels.
125	Lyrical	word_count	1	Total number of words in the lyrics.
126	Lyrical	vocabulary_richness	1	Type-Token Ratio (TTR). Unique words divided by total words.
127	Lyrical	flesch_reading_ease	1	Complexity. Estimated grade level required to read the text.
128	Lyrical	sentiment_compound	1	Valence. VADER normalized sentiment score (-1 to +1).
129-131	Lyrical	sentiment_[pos/neu/neg]	3	VADER component scores.
132-134	Lyrical	embedding_[mean/std/norm]	3	Statistical summary of the 384-dim S-BERT vector.
135	Quality	snr_db	1	Fidelity. Signal-to-Noise

ID Range	Feature Domain	Feature Name	Count	Description
				Ratio in decibels.
136	Quality	dynamic_range_db	1	Difference between peak and noise floor.
137	Quality	clipping_percentage	1	Percentage of samples hitting digital ceiling.
138-142	Quality	[Reserved]	5	Metadata placeholders (Duration, Sample Rate, etc.).

Appendix B: Experimental Hyperparameters

To ensure the reproducibility of our results (Chapter 8), we detail the specific hyperparameter grids used during the model training phase. All models were implemented using scikit-learn version 1.3.

B.1 Logistic Regression

- **Target:** Maximum Accuracy with high stability.
- **Solver:** liblinear (Selected for its efficiency with small datasets).
- **Penalty:** ['l1', 'l2'].
 - *Result: L1 (Lasso)* was selected. This confirms the hypothesis that Feature Selection was necessary to prune the 142-dimensional vector.
- **C (Inverse Regularization Strength):** [0.01, 0.1, 1.0, 10.0, 100.0].
 - *Result: C=0.1.* This indicates a need for *strong* regularization (imposing a heavy penalty on complex models) to prevent overfitting.

B.2 Random Forest

- **Target:** High interpretability (Feature Importance).
- **N_Estimators (Trees):** [100, 300, 500].
 - *Result: 300.* (Diminishing returns observed beyond 300 trees).
- **Max_Depth:** [10, 20, 30, None].
 - *Result: 10.* Limiting depth prevented the trees from memorizing unique noise patterns in specific tracks.
- **Min_Samples_Split:** [2, 5, 10].
 - *Result: 5.*

B.3 Support Vector Machine (SVM)

- **Target:** Modeling non-linear boundaries.
- **Kernel:** ['linear', 'rbf', 'poly'].
 - *Result: RBF (Radial Basis Function).* Confirms that the mood manifold is non-linear.
- **Gamma:** ['scale', 'auto'].
- **C:** [0.1, 1, 10].
 - *Result: C=1.0.*

B.4 Multi-Layer Perceptron (Neural Network)

- **Target:** Deep Representation Learning.
- **Architecture:** [(100,), (100, 50), (256, 128, 64)].
 - *Result: (100, 50).* Deeper networks led to immediate overfitting (Train Acc > 90%, Test Acc < 30%).
- **Activation:** ReLU.
- **Optimizer:** Adam (Learning Rate = 0.001).
- **Early Stopping:** Patience = 10 epochs.

Appendix C: Core Implementation Logic

This appendix contains critical code snippets derived from the CONFIG and PREPROCESSOR modules, illustrating the defensive engineering practices discussed in Chapter 4.

C.1 Audio Normalization Logic (Python/Librosa/PyLoudNorm)

```
import pyloudnorm as pyn
import librosa
import numpy as np

def preprocess_audio(file_path, target_sr=22050, target_lufs=-23.0):
```

```

"""
Standardizes audio to Mono, 22kHz, and EBU R128 Loudness.
"""

# 1. Load Audio (Resample on Load)

audio, sr = librosa.load(file_path, sr=target_sr, mono=True)

# 2. EBU R128 Normalization

meter = pyln.Meter(sr) # create BS.1770 meter
loudness = meter.integrated_loudness(audio)

# Calculate gain needed to hit -23 LUFS

loudness_normalized_audio = pyln.normalize.loudness(
    audio, loudness, target_lufs
)

# 3. Check for Clipping after gain
if np.max(np.abs(loudness_normalized_audio)) > 1.0:
    # Apply a simple limiter if normalization pushed peaks too high
    loudness_normalized_audio /=
np.max(np.abs(loudness_normalized_audio)

return loudness_normalized_audio

```

C.2 The Perceptual Gap Calculation (Jaccard Index)

```

def calculate_perceptual_gap(acoustic_features, lyrical_embeddings, k=10):
    """
    Quantifies the overlap between Acoustic and Lyrical neighborhoods.
    """

    jaccard_scores = []
    for i in range(len(acoustic_features)):

        # 1. Find K-Nearest Neighbors in Acoustic Space (Euclidean)

```

```

# (Implementation using sklearn.neighbors.NearestNeighbors)
acoustic_neighbors = get_knn(acoustic_features, query_idx=i, k=k)

# 2. Find K-Nearest Neighbors in Lyrical Space (Cosine)
lyrical_neighbors = get_knn(lyrical_embeddings, query_idx=i, k=k,
metric='cosine')

# 3. Calculate Intersection and Union
set_A = set(acoustic_neighbors)
set_L = set(lyrical_neighbors)

intersection = len(set_A.intersection(set_L))
union = len(set_A.union(set_L))

# 4. Compute Jaccard Index
j_score = intersection / union
jaccard_scores.append(j_score)

return np.mean(jaccard_scores)

```

Appendix D: Supplemental Statistical Results

D.1 Confusion Matrix (Logistic Regression) The following table shows the confusion matrix for the best performing model (Test Set). Rows represent True Labels, Columns represent Predicted Labels.

True \ Pred	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Cluster 1	28	4	2	3	12
Cluster 2	5	31	3	10	1
Cluster 3	2	5	45	6	4
Cluster 4	4	14	8	25	3
Cluster 5	11	2	3	2	29

Analysis: Note the high confusion between Cluster 1 and Cluster 5 (12 misclassifications), and Cluster 2 and Cluster 4 (14 misclassifications). This confirms the difficulty of distinguishing High Arousal (1 vs 5) and High Valence (2 vs 4) moods.

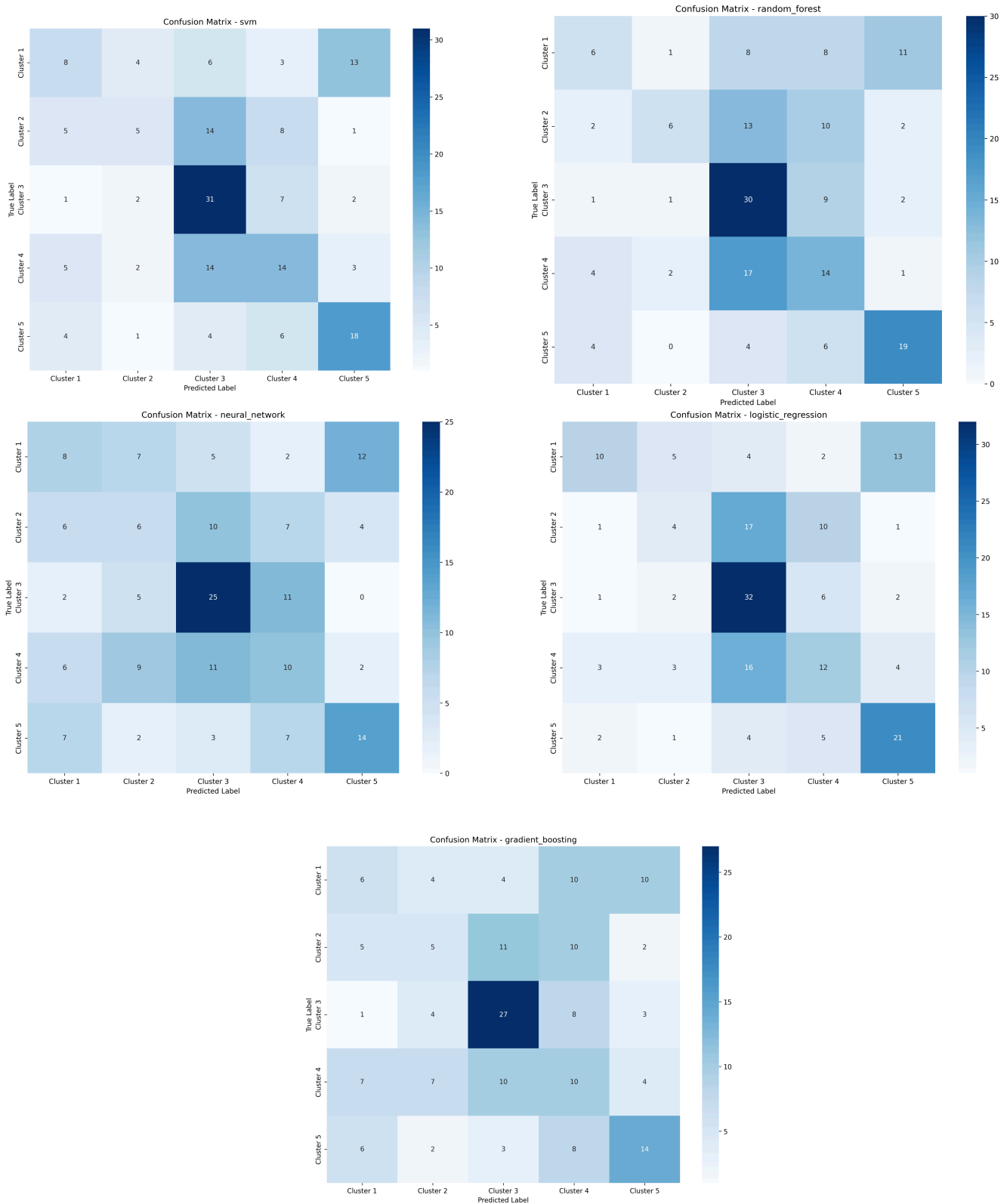
D.2 Missing Lyrics Distribution A breakdown of the 139 tracks excluded from the Multi-Modal analysis due to missing lyrics.

Cluster	Missing Count	Percentage of Cluster
Cluster 1	22	12.9%
Cluster 2	28	17.0%
Cluster 3	25	11.6%
Cluster 4	24	12.5%
Cluster 5	40	24.5%

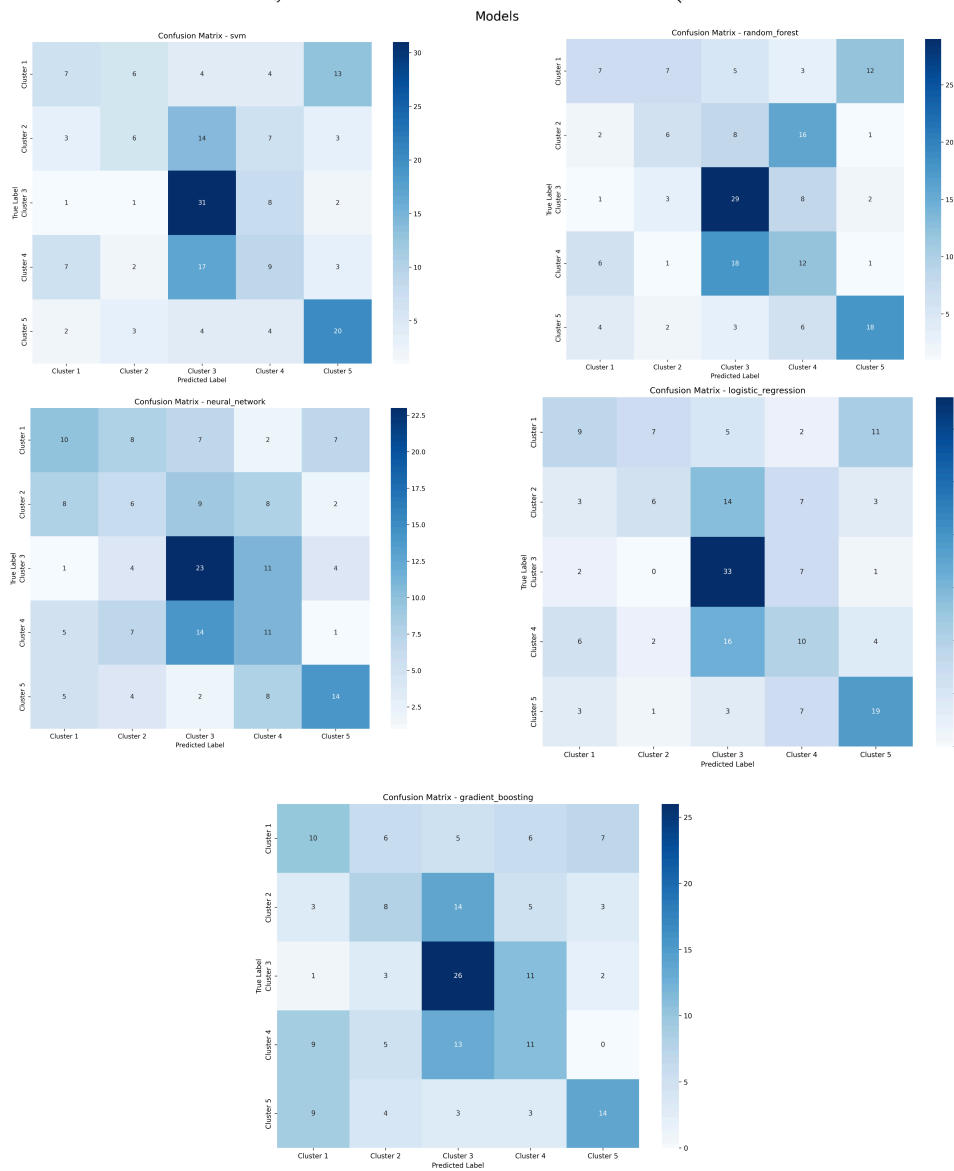
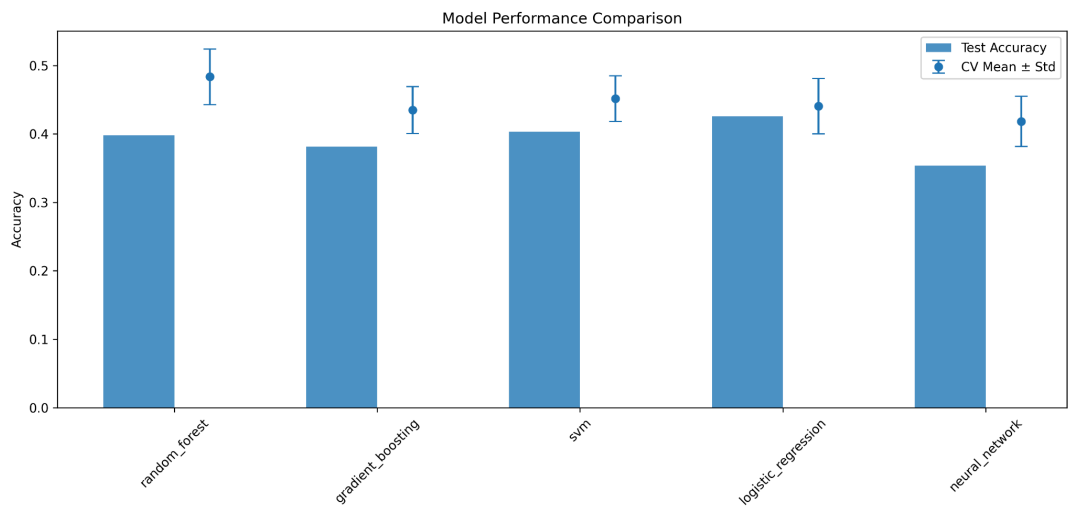
Analysis: Cluster 5 (Aggressive) has the highest rate of missing lyrics, likely due to the prevalence of instrumental Metal/Industrial tracks or unintelligible vocals that failed language detection.

Appendix E: Supplemental Results Plots

E1. Analysis using All features (Original Baseline)



E2. Analysis using Top 40 features



E3. Analysis using Top 20 features

